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ABSTRACT

This report outlines the development process of the **FAYDH** system, a unified digital solution designed to manage and optimize in-kind donations (such as food, clothing, and furniture) across charitable organizations. Current donation processes frequently suffer from fragmentation across different categories and a lack of real-time transparency and tracking, which compromises efficiency. The **FAYDH** project introduces an integrated mobile and web application that automates and enhances the entire donation workflow. By leveraging real-time tracking mechanisms, code verification for delivery, and smart matching algorithms, the system ensures that donations are safely and efficiently transferred. **FAYDH** minimizes resource waste, enhances donor trust, and delivers a faster, more organized charitable experience, supporting the digital transformation of the non-profit sector.

Table of Contents

1. INTRODUCTION	1
2. DOMAIN ANALYSIS	2
3. RISK/CONSTRAINTS	7
4. PROJECT PLAN	8
5. QUALITY ASSURANCE PLAN	9
6. REQUIREMENTS	10
6.1 Requirement Elicitation	10
6.2 FUNCTIONAL REQUIREMENTS	11
6.3 NON-FUNCTIONAL REQUIREMENTS	16
6.4 Design Constraint	16
7. PROBLEM COMPLEXITY	17
8. SYSTEM USE CASES	18
8.1 Use Case Diagram	18
8.1.1 Donor Use Case Diagram	18
8.1.2 Charity Use Case Diagram	19
8.1.3 Driver Use Case Diagram	20
8.1.4 Administrator Use Case Diagram	21
8.1.5 Full Use case	22
8.2 Use Case Descriptions	23
8.2.1 Create Donation Post Use Case description	23
8.2.2 Choose Pickup Available Time Use Case Description	25
8.2.3 Join chat Use Case Description	26
8.2.4 Reserve donation post Use case description	27
8.2.5 Cancel donation reservation post Use case description	29
8.2.6 View available donation post details Use Case description	30
8.2.7 Choose Available Time Slots for Delivery Use Case Description	31
8.2.8 Request to join charity Use Case Description	32
8.2.9 Update delivery status Use Case description	33
8.2.10 Sign Up charity Use Case Description	34
9. ANALYSIS CLASS	35
9.1 Create donation post	35
9.2 Choose pickup available time	35
9.3 Join chat	36
9.4 Reserve donation post	36
9.5 Cancel donation reservation post	37
9.6 View available donation post details	37
9.7 Choose available time slots for delivery	38
9.8 Request to join charity	39
9.9 Update delivery status	39

9.10 Sign up charity	39
9.11 Full analysis class	40
10. INTERACTION DIAGRAM	41
10.1 Create Donation Post	42
10.2 Choose pickup available time	43
10.3 Join chat	44
10.4 Reserve donation post	45
10.5 Cancel donation reservation post	46
10.6 View available donation post details	47
10.7 Choose available time slots for delivery	48
10.8 Request to join charity	49
10.9 Update delivery status	50
10.10 Sign Up charity	51
11. DESIGN CLASS	52
12. SYSTEM ARCHITECTURE	53
12.1 MVVM Architecture style	53
12.2 MVVM Architecture for Admin Website	55
12.3 Architecture Constraints	55
12.4 MVVM Architecture compared to other Architectures	55
13. User Interface Mockup	57
13.1 Welcome Page & Sign In	57
13.2 Sign Up for charity	58
13.3 Create Donation Post	58
13.4 Join Chat	60
13.5 Choose Available Pick-up time	61
13.6 Reserve donation post	62
13.7 Cancel Donation Reservations	63
13.8 Accept/Reject Drivers' Request	63
13.9 Update delivery status	64
13.10 Driver Request to join	67
14. DATABASE	68
14.1 Database Schema	68
15. ALGORITHMS	70
15.1 delivery tasks Haversine	70
16. EXPECTED DEPLOYMENT	71
17. TEST SCENARIO	72
17.1 Create donation post test cases	72
17.1.1 Invalid scenario	72
17.1.2 Valid scenario	72
17.2 Choose pickup available time test case	73
17.2.1 Valid scenario	73
17.3 Join chat test cases	74
17.3.1 Valid scenario	74
17.3.2 Invalid scenario	74

17.4 Reserve donation post test cases	75
17.4.1 Valid scenario	75
17.4.2 Invalid scenario	75
17.5 Cancel donation reservation post	76
17.5.1 Valid scenario	76
17.5.2 Invalid scenario	77
17.6 view available donation post details	78
17.6.1 valid scenario	78
17.6.2 Invalid scenario	78
17.6.3 Invalid scenario	79
17.6.4 Invalid scenario	79
17.7 Choose available time slots for delivery	80
17.7.1 valid scenario	80
17.7.2 invalid scenario	80
17.8 Request to join charity	81
17.8.1 Valid scenario	81
17.8.1 Invalid scenario	82
17.9 Update delivery status	82
17.9.1 Valid scenario	82
17.9.2 Invalid scenario	83
17.10 Sign Up charity	84
17.10.1 Valid Sign up	84
17.10.2 Invalid Email Sign up	84
17.10.3 Invalid Password Sign up	85
17.10.4 Invalid Phone number Sign up	86
17.10.5 Invalid license number Sign up	86
18. PROJECT STATUS	87
19.Future Work	89
20. CONCLUSION	90
21. REFERENCE	91
22.Appendix	93
Appendix 1	93
Appendix 2	94
Appendix 3	99
Appendix 4	100

List of Figures

Figure 1. Kasa platform	2
Figure 2. Khiyrat platform	2
Figure 3. Al-Bir charitable society	3
Figure 4. Al-Inayah association	3
Figure 5. Irgah charitable association	4
Figure 6. Tadweer charitable association	4
Figure 7. Project plan	8
Figure 8. Donor use case diagram	18
Figure 9. Charity use case diagram	19
Figure 10. Driver use case diagram	20
Figure 11. Admin use case diagram	21
Figure 12. Full use case diagram	22
Figure 13. Create donation post VOPC	35
Figure 14. Choose pickup available time VOPC	36
Figure 15. Join chat VOPC	36
Figure 16. Reserve donation VOPC	37
Figure 17. Cancel donation reservation VOPC	37
Figure 18. View available donation post details VOPC	38
Figure 19. Choose available time slots for delivery VOPC	38
Figure 20. Request to join charity VOPC	39
Figure 21. Update delivery status VOPC	39
Figure 22. Sign up charity VOPC	40
Figure 23. Full analysis class	41
Figure 24. Create donation post sequence diagram	42
Figure 25. Choose pickup available time sequences diagram	43
Figure 26. Join chat sequences diagram	44
Figure 27. Reserve donation post sequences diagram	45
Figure 28. Cancel donation reservation post sequences diagram	46
Figure 29. View available donation post details sequences diagram	47
Figure 30. Choose available time slots for delivery sequences diagram	48
Figure 31. Request to join charity sequences diagram	49
Figure 32. Update delivery status sequences diagram	50
Figure 33. Sign up charity sequences diagram	51
Figure 34. FAYDH design class	52
Figure 35. FAYDH Architecture Diagram	56
Figure 36. Welcome and sign in page	58
Figure 37. Sign up for charity	58
Figure 38. Donor Homepage, Donor clicks on New Donation to create a donation post.	58
Figure 39. The donor fills in a form with the relevant type	59

Figure 40. The donor can choose an existing location or add a new Location, after confirming the donation post the message will be received by who(Driver)	59
Figure 41. The donor can join a chat with the charity to give them more details about the donation.	60
Figure 42. The donor can choose the available time for delivery pick up	61
Figure 43. Confirmation steps and view the donation post details with scheduled pickup	61
Figure 44. Charity Home page and receive notification for donation post, charity can view details and reserve a donation post and publish the time for donation pick- up	62
Figure 45. Confirmation message for reserved donation post	62
Figure 46. Charity can cancel reserved donation	63
Figure 47. Charity view driver request and can accept or reject them	64
Figure 48. Driver mark himself to available to get assigned a delivery from the system	65
Figure 49. Driver can view active delivery and delivery history	65
Figure 50. Driver view delivery details and enters the verification code	65
Figure 51. Driver view and navigate to Drop-off location,and receive notification.	66
Figure 52. Driver request to join charity	67
Figure 53. Conformation message for requested successfully	67
Figure 54. FAYDH database schema	70
Figure 55. FAYDH Algorithm	71
Figure 56. FAYDH deployment diagram	72
Figure 57. FAYDH Questionnaire	93
Figure 58. FAYDH Questionnaire Result	95
Figure 59. FAYDH Interview Question	99
Figure 60 . FAYDH Interview Answer	100

List of Tables

Table 1.Proposed systems vs. existing applications	6
Table 2.Risks	8
Table 3.Create Donation Post use case description	24
Table 4.Choose Pickup Available Timet use case description	26
Table 5.Join chat use case description	27
Table 6. Reserve donor post use case description	28
Table 7.Cancel donation reservation post use case description	30
Table 8.View available donation post details use case description	31
Table 9.Choose Available Time Slots for Delivery use case description	32
Table 10.Request to join charity use case description	33
Table 11.Update delivery status use case description	34
Table 12.Sign up charity use case description	35
Table 13.Create donation post test case	72
Table 14.Create donation post test case	73
Table 15.Choose pickup available time test case	73
Table 16.Join chat test case	74
Table 17.Join chat test case	75
Table 18.Reserve donation post test case	75
Table 19.Reserve donation post test case	76
Table 20.Cancel donation reservation post test case	77
Table 21.Cancel donation reservation post test case	77
Table 22.View available donation post details test case	78
Table 23.View available donation post details test case	78
Table 24.View available donation post details test case	79
Table 25.View available donation post details test case	80
Table 26.Choose available time slots for delivery test case	80
Table 27.Choose available time slots for delivery test case	81
Table 28.Request to join test case	81
Table 29.Request to join test case	82
Table 30.Update delivery status test case	83
Table 31.Update delivery status test case	83
Table 32.Sign up test case	84
Table 32.Sign up test case	85
Table 33.Sign up test case	85
Table 34.Sign up test case	86
Table 35.Sign up test case	87
Table 36.Project status	89

1. Introduction

● Problem

The charitable giving ecosystem in the Kingdom of Saudi Arabia and across the Arab world is currently fragmented, with multiple digital platforms and channels through which donors can contribute. While platforms such as *Ehsan* have become prominent references for digital philanthropy, they predominantly facilitate **monetary donations only**, without enabling donors to provide direct, in-kind support such as food, clothing, furniture, or other essential services that meet the immediate needs of beneficiaries.

so there is no unified platform that allows donors to contribute multiple types of donations simultaneously, such as combining monetary contributions with in-kind goods or service-based support within a single, integrated donation process. This lack of multi-donation integration represents a core challenge, as donors are often required to use separate platforms or channels for different types of contributions, increasing complexity and reducing overall engagement.

This limitation restricts donors' ability to provide tangible assistance, leaving gaps in both the efficiency and impact of charitable contributions. Official statistics indicate that the *Ehsan* platform has collected over **10 billion SAR** in monetary donations since its launch, benefiting more than **4.8 million individuals** through millions of transactions. Despite these impressive figures, the focus on financial contributions alone leaves a critical gap for donors who wish to provide **direct, tangible support**, while the multiplicity of platforms complicates the donation process, limiting the donors' ability to compare options, verify beneficiaries' needs, and ensure effective use of contributions. This fragmentation reduces overall efficiency, increases redundancy, and diminishes public trust in the charitable system.

● Proposed Solution

To address these challenges, the **FAYDH** system offers a unified digital platform that consolidates all certified charitable organizations into a single interface, providing an innovative solution that focuses exclusively on in-kind donations and direct services rather than monetary contributions. Donors can contribute essential items such as food, clothing and furniture. All contributions are delivered through authorized charities and designated drivers, ensuring that aid reaches the intended recipients efficiently, transparently, and reliably. By centralizing participating organizations and clearly presenting all available in-kind donation options, **FAYDH** reduces the time and effort required from donors, enables informed decision-making, and maximizes the impact of charitable contributions.

Furthermore, the **FAYDH** system aligns closely with the objectives of **Saudi Vision 2030** by promoting sustainable social development and digital innovation. The platform supports a culture of responsible and sustainable giving through trusted digital channels, increases the operational efficiency of charitable work through technology-enabled direct delivery of aid, strengthens transparency and accountability in donation management in line with national governance standards, and empowers communities to actively engage in social and humanitarian initiatives. In this way, **FAYDH** serves as a single unified platform that brings together all charities, simplifying the donation process and making it more technologically advanced. By combining direct, in-kind donations with transparency, speed, and digital innovation, the platform ensures measurable impact

for beneficiaries while enhancing community participation in social development, fully aligning with the objectives of Vision 2030.

2. Domain Analysis

Individual donations in Saudi Arabia still face challenges, particularly when it comes to in-kind contributions such as food, clothing, or flowers from social events. Although several platforms and initiatives exist, most focus on specific types of donations or rely on traditional methods, leaving gaps in integration, real-time tracking, and user engagement. This highlights the need for a comprehensive and intelligent system that unifies the donation process and provides efficiency, transparency, and sustainability. [1]

Existing Solutions

- **Kasa platform**



Figure 1. Kasa platform

Kasa is a project that collects used clothing and delivers it to partner charities. These organizations then sort and recycle the clothes in a sustainable and hygienic way, and collect used clothing to sort and sustainably recycle it.[2]

- **Khiyrat platform**



Figure 2.Khiyrat platform

This is a non-profit organization established in 2015 that specializes in preserving surplus food from banquets held at hotels, restaurants, palaces, resorts, and homes. A specialized field team sorts and repackages this surplus food before distributing it to families in need.[3]

- **Al-Bir Charitable Society**



Figure 3. Al-Bir charitable society

The Al-Bir Charitable Society in Riyadh is one of the oldest and most well-known charitable organizations in the Kingdom. It provides a variety of services, including receiving financial and in-kind donations, such as clothing and food, in addition to programs to sponsor orphans and support needy families. What distinguishes it is its high credibility, given its long history and closeness to the local community through its widespread branches and centers.[4]

- **Al-Inayah association**



Figure 4. Al-Inayah association

Al-Inayah Association is a non-profit organization dedicated to improving the lives of individuals and families. It collects clean, used clothing for reuse and sells damaged clothing to generate income. This income is then used to provide shopping vouchers to beneficiaries, which preserves their dignity and gives them the freedom to choose what they need. The association also promotes a culture of reducing waste and builds effective community partnerships with businesses, donors, and philanthropists to achieve sustainable development.[5]

- **Irqah charitable association**



Figure 5.Irqah charitable association

Irqah Charitable Association is one of the leading organizations in Riyadh that collects furniture and distributes it to families in need. You can visit their website or contact them directly to arrange a donation. It is a non-profit organization located in Riyadh that provides financial and in-kind assistance to families in need. It was founded in 2009 by Sheikh Mohammed Al-Rawwaf (may he rest in peace) and operates under the supervision of the National Center for the Development of the Non-Profit Sector.[6]

- **Tadweer charitable association**



Figure 6.Tadweer charitable association

Tadweer Association is a non-profit organization dedicated to promoting sustainability and reducing waste through recycling and community awareness. Its vision is to lead in recycling through a sustainable work system, ensuring that surplus food and resources are reused safely and effectively. Tadweer also educates the community on responsible consumption and environmental preservation, contributing to a more sustainable future across the Kingdom.[7]

Table below presents a comparison of existing donation platforms listed above and the proposed system based on their key features and functionalities.

Feature/Platform	Kesaa	Khiyrat	Al-Bir	Al-Inayah	Irqah	Tadweer	Proposed System
In-kind donations (food, clothes, flowers, furniture)	✓ clothes	✓ food	✓ food, clothes	✓ clothes	✓ furniture	✓ food	✓ all types
Real-time tracking of donations	✗	✗	✗	✗	✗	✗	✓
Seasonal / campaign management	✗	✓ food drives	✓ traditional	✗	✗	✓ "Naqa'a-Ramadan Campaign"	✓
Integration with delivery / verification	✗	✗	✗	✗	✗	✗	✓
Donor engagement (badges, certificates, gamification)	✗	✗	✗	✗	✗	✗	✓
AI technologies (image recognition, predictive analytics, smart matching)	✗	✗	✗	✗	✗	✗	✓
Bilingual support (Arabic/English)	✗	✗	✗	✗	✗	✓	✓
Reporting & analytics	✗	Limited	Limited manual reports	Limited (vouchers)	Limited	Limited manual reports	✓ real-time dashboard & smart reports)

Table 1.Proposed systems vs. existing applications

The proposed system supports multiple user categories, including donors, drivers, charities, each with clearly defined roles to ensure an integrated and transparent donation workflow. Donors can submit donation posts and schedule pickups or drop-offs, which are verified using a verification code to ensure the authenticity and traceability of in-kind donations. Drivers are responsible for collecting and delivering verified donations, while Charities reserve and receive in-kind donations.. Unlike existing charity platforms in Saudi Arabia, which primarily focus on financial donations and offer limited mechanisms for verifying in-kind contributions, the proposed system introduces code verification at key process stages (donation submission, pickup, and delivery). This technical approach directly addresses trust and verification challenges identified in the domain study by preventing duplicate claims, confirming donation handovers, and ensuring that donations reach the intended charities. As a result, the system enhances transparency, reduces manual verification errors, and improves operational efficiency and user trust.

3. Risk/Constraints

Most enterprises ignore the risks that are associated with any stages of the software development life cycle (SDLC). Risks and constraints must be analyzed otherwise they will create unforeseen challenges. In our project, we will go through different stages of the SDLC where risks are encountered. For that reason, we created a risk analysis table to try avoiding these risks.

NO.	Risk	Type	Severity	Likelihood	Risk Management Strategy
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1	Complexity in third-party API integration	Technical	High	High	Use open-source libraries, conduct thorough compatibility testing, and develop a clear API integration plan.
2	Team communication and coordination breakdown	Organizational	High	Medium	Establish a clear project management tool (e.g., Trello, Asana), define roles and responsibilities, and schedule regular team meetings.
3	Timeline and deadline management	Organizational	High	Medium	Break the project into smaller, manageable milestones, and create a detailed Gantt chart or project schedule.
4	Budget limitations restricting tools and services	Financial	High	High	Rely on open-source frameworks and free APIs, and explore alternative, cost-effective solutions.
5	Compliance with food safety and local charity laws	Legal/Policy	High	High	Consult with local authorities or legal experts and incorporate a compliance checklist into the project plan.
6	Data Privacy Mitigation	Technical	High	High	Donor address/location data is stored in Firebase Firestore, which provides managed encryption at rest by default. Additionally, access to this data is controlled using Firestore Security Rules (role-based authorization) to ensure only authorized roles (e.g., the assigned driver and relevant charity) can read it.
7	Data security breaches	Technical	High	Medium	Implement robust data encryption protocols, adhere to best practices for data handling, and conduct regular security audits.
8	Performance issues due to limited hosting resources	Infrastructure	Medium	High	Optimize the application code for efficiency, and use a reliable, scalable cloud hosting service that offers a free tier.

9	Cross-platform compatibility issues during testing	Technical	Medium	High	Use a cross-platform development framework (e.g., Flutter, React Native), and conduct extensive testing on various devices and operating systems.
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Table 2.Risks

4. Project Plan

A **project plan**, also known as a **project management plan**, is a document that outlines how a project will be executed, monitored, controlled, and closed. It acts as a roadmap that visualizes milestones, activities, timelines, and task dependencies—commonly represented through **Gantt charts** that illustrate the project’s progress from initiation to completion using Microsoft’s project plan.

For this project, **FAYDH** will follow a **structured, phase-based development approach** aligned with the **Waterfall methodology**, as the **Graduation Project Committee (GPC)** has predetermined the project’s deadlines and milestones. This ensures that each stage is completed and reviewed before proceeding to the next, providing clarity, traceability, and organized progress.

The project plan establishes the foundation for **FAYDH’s** software development process. It begins with defining objectives, conducting domain research, identifying potential risks, and outlining a **quality assurance strategy**.

Subsequent phases focus on **system analysis and design**, including the specification of functional and non-functional requirements, complexity assessment, and the creation of **use case diagrams, class diagrams, and system architecture**. It also involves designing **user interface mockups**, developing **database schemas**, and defining **algorithms, testing procedures, and deployment plans**.

Overall, this plan ensures a smooth and systematic transition from initial planning to final implementation, maintaining consistency with the structured workflow defined by the project’s methodology.

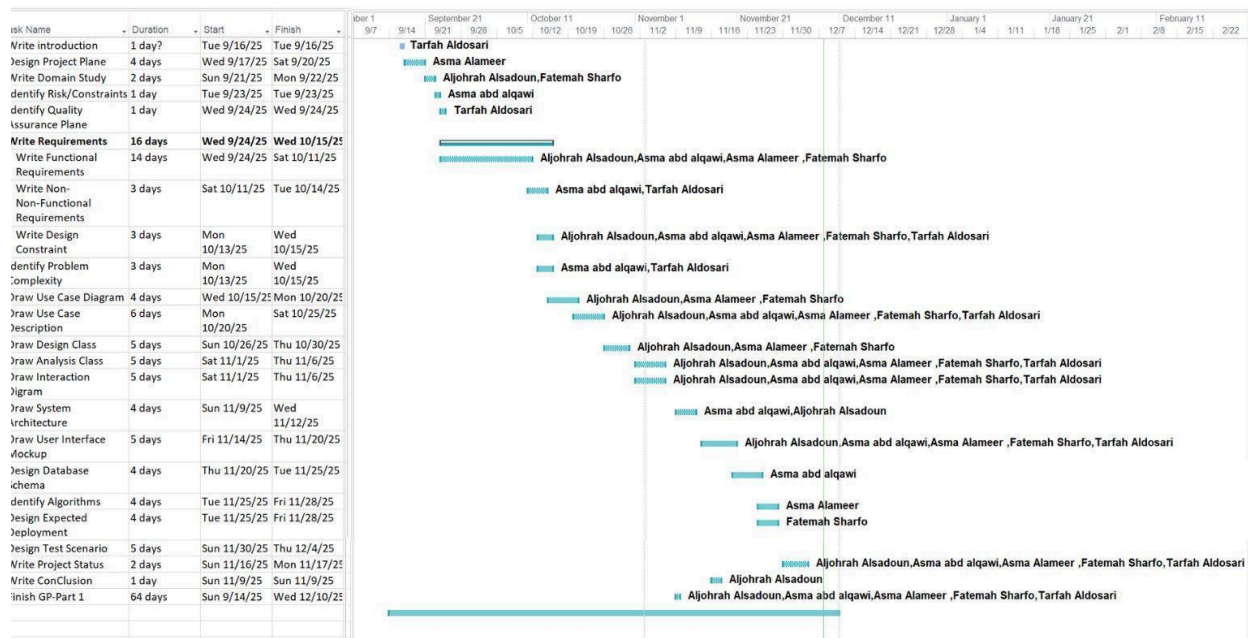


Figure 7. Project plan

5. Quality Assurance Plan

This section outlines the activities we will perform during the project lifecycle to ensure that the system meets high standards of quality, reliability, and transparency.

A. Formal Reviews

Should follow an inspection and should be used at the end of each major phase and/or iteration (Requirements specs, design, test plan, etc.). Formal reviews should be conducted in a formal way with the presence of students and the adviser (and eventually the exam committee). There must be a written report documenting all findings from the formal review session. The adviser is responsible to follow up with students. The report must be filled by the adviser and eventually the member of the exam committee.

B. Inspections

Inspections will be performed at the team level as a structured peer-review activity. Each artifact (e.g., requirements documents, UI mock-ups, database schema, and code modules) will be examined systematically to detect defects, inconsistencies, and non-compliance with project standards. This process ensures early detection of issues, reduces rework, and increases the reliability of the app.

C. Training Team Members

Each team member will strengthen their technical and domain knowledge through online courses and tutorials relevant to the project's technologies. Training will focus on:

- **Mobile and Web Development** (Flutter/Dart, or equivalent frameworks).
- **Database Design & Security** (SQL, secure authentication, encryption techniques).
- **Real-time Data & Reporting** (APIs, dashboards, data visualization).

- **Charity and Donation Systems Best Practices** to align with humanitarian values and regulatory requirements.

D. Testing & Validation

A systematic testing process will be applied:

- **Unit Testing** for individual modules (e.g., login, reporting, dashboard).
- **Integration Testing** to verify interactions between system components (e.g., donor → platform → charity → beneficiary).
- **System Testing** to ensure the platform functions as intended under real-world conditions.
- **User Acceptance Testing (UAT)** with simulated donors and charity staff to confirm usability, transparency, and trust in the donation process.

E. Walkthrough

Walkthrough is an informal type of review to evaluate the documentation and code without planning for that. The team members will give comments on the work and change any parts based on these comments [8].

F. User Acceptance Testing (UAT)

The system will be tested with actual charity members to ensure it meets real-world needs. Participants will perform key tasks such as viewing donation posts, reserving donations, receiving notifications, and using the dashboard. Feedback on usability, interface clarity, and trust in the system will be collected and documented in a formal UAT report. The team will address any issues identified to improve the system before deployment.

6. Requirements

The **requirements phase** is the second stage of the software development life cycle. It defines the features and functionalities that the system will provide to its stakeholders. For **FAYDH**, this phase focuses on identifying and documenting what the system should do and how it should perform.

There are two main categories of requirements: **functional requirements**, which specify the system's behaviors and services, and **non-functional requirements**, which describe the quality attributes such as performance, usability, and security.

6.1 Requirement Elicitation

To accurately identify and define the system requirements for the **FAYDH** smart donation management platform, multiple requirement elicitation techniques were employed, including questionnaires, interviews, and domain analysis. The integration of these methods ensured a comprehensive understanding of user needs, technical constraints, and existing gaps in the current donation processes.

The **questionnaire** was designed to gather insights from both donors and beneficiaries regarding their experiences with charitable giving and receiving. It explored several aspects such as preferred donation types, current donation methods, challenges in the delivery or receipt process, and

attitudes toward adopting smart solutions like mobile applications, real-time notifications, and AI-based recommendations. The collected responses revealed a strong demand for more transparent, efficient, and user-friendly donation management systems. Participants emphasized the importance of trust, real-time tracking, and simplified logistics to enhance their overall experience.

In parallel, **interviews** were conducted with representatives from various charitable organizations. These sessions provided a deeper understanding of the operational structures, technological capabilities, and data management practices of existing charities. The interviews also uncovered practical challenges, such as limited digital integration and inefficiencies in managing beneficiary information.

Furthermore, **domain analysis** was performed by examining existing charitable applications with similar objectives. This analysis highlighted their strengths and limitations, enabling the identification of improvement opportunities. Based on these findings, the proposed **FAYDH** application was designed to integrate and enhance successful features while introducing new functionalities aimed at addressing the identified challenges.

Through this systematic approach combining user input, expert insights, and comparative analysis, the development process ensured that the **FAYDH** platform not only meets but exceeds stakeholder expectations. The outcome is a user-centered solution that simplifies the donation process, strengthens transparency, and fosters effective collaboration between donors, beneficiaries, and charitable organizations.

6.2 Functional Requirements

- **Definitions:**

Donation Post: refers to a post created by a donor to offer clothes, food, furniture, or flowers for donation. The term “Donation Post” is used instead of “Donation Request” because the donor is offering a donation to all charities, rather than requesting a specific charity.

- **Donor Requirements**

BR1. The donor shall be able to sign up using first name, last name, phone number, email, password, profile picture(optional), and location.

BR2. The donor shall be able to sign in using email and password.

BR3. The donor shall be able to sign out.

BR4. The donor shall be able to reset their password using an OTP sent to their email.

BR5. The donor shall be able to view their profile information that includes first name, last name, phone number, location, email, and profile picture.

BR6. The donor shall be able to edit their profile information that includes first name, last name, picture, phone number, email, and location.

BR7. The donor shall be able to view a list of all charities.

BR8. The donor shall be able to search for charity.

BR9. The donor shall be able to view the charities' information, including the charity's name, brief description, profile picture, policies, email, phone number, location, and working times.

BR10. The donor shall be able to create a donation post.

BR11. The donor shall be able to select the type of donation that includes (food, clothes, flowers, and furniture.

BR12 The donor shall be able to fill out a food donation form that includes Food item name, Quantity, photo(s), food description.

BR13 The donor shall be able to fill out a clothes donation form with title , clothing type (men, women, children, unisex), condition (new, gently used, used), season (summer, winter, all seasons), category (shirt, pants, jacket, shoes, other), and photo(s).

BR14. The donor shall be able to fill out a flowers donation form that includes title, flower type (roses, tulips, mixed bouquet, artificial), quantity / number of bouquets, occasion (wedding, funeral, general), freshness level (fresh, dried, artificial), photo(s).

BR15. The donor shall be able to fill in a furniture donation form that includes title, furniture type, condition (new, gently used, used), photo(s).

BR16. The donor shall be able to add a new location when creating a new donation post.

BR17. The donor shall be able to select the location from the saved location list when they create a new donation post.

BR18. The donor shall be able to view the list of their donation posts.

BR19. The donor shall be able to view the status of their submitted donation posts if they are (available, reserved, delivered, expired).

BR20. The donor shall be able to view the charity's information on who has reserved their post, including name of charity, phone number, email, available date and time slots for pick-up and profile picture.

BR21. The donor shall be able to choose the Pick-up available date and time after the charity reserves their post.

BR22. The donor shall be able to edit a donation post before the charity reserves it.

BR23. The donor shall be able to delete a donation post before the charity reserves it.

BR24. The donor shall be able to track the driver's location while waiting for donation pickup.

BR25. The donor shall be able to view the verification code to confirm the delivery from Whatsapp.

BR26. The donor shall be able to join a chat session with charity after their donation post has been reserved by a charity.

BR27. The donor shall be able to receive a notification when the charity reserves their donation post.

BR 28. The donor shall be able to receive diamond reward when posting up to 100 donation posts.

BR 29. The donor shall be able to receive gold reward when posting up to 50 donation posts.

BR 30. The donor shall be able to receive bronze reward when posting up to 20 donation posts.

BR31. The donor shall be able to receive a notification when the charity cancels the donation post-reservation.

- **Charity Requirements**

BR32. The charity shall be able to sign up using the charity name, license number, profile picture(optional), phone number, email, password, brief description, policies, working times, and location.

BR33. The charity shall be able to sign in using email and password.

BR34. The charity shall be able to sign out.

BR35. The charity shall be able to reset their password using an OTP sent to their email.

BR36. The charity shall be able to view its profile information, including charity name, license number, profile picture, phone number, email, password, brief description, policies, working times and location.

BR37. The charity shall be able to edit its profile information, including charity name, license number, profile picture, phone number, email, password, brief description, policies, working times

and location.

BR38. The charity shall be able to view a dashboard that includes the total number of donations received, and number of driver requests, and visual charts for donation trends over time.

BR39. The charity shall be able to view a list of available donation posts.

BR40. The charity shall be able to view donation post details, including donor name, phone number, email, donation type, and the donation form relevant to the donation type.

BR41. The charity shall be able to reserve a donation post from the donor.

BR42. The charity shall be able to view a list of their reserved donations posts.

BR43. The charity shall be able to view reservation donation post details

BR44. The charity shall be able to choose the available date and time slots for delivery to the donor while the post is reserved.

BR45. The charity shall be able to edit the published time slots before the donor selects pick-up date and time.

BR46. The charity shall be able to start a chat session with the donor after the post is reserved.

BR47. The charity shall be able to cancel the reservation of a donation post.

BR48. The charity shall be able to view a list of driver requests.

BR49. The charity shall be able to accept drivers' requests.

BR50. The charity shall be able to reject drivers' requests.

BR51. The charity shall be able to receive a notification when the donor submits the post.

BR52. The charity shall be able to receive a notification when the status of the donation post turns from reserved to available.

- **Admin Requirements**

BR53. The admin shall be able to sign in.

BR54. The admin shall be able to sign out.

BR55. The admin shall be able to reset the password using an OTP sent to his/ her email.

- BR56. The admin shall be able to view a list of charity requests.
- BR57. The admin shall be able to accept charity requests.
- BR58. The admin shall be able to reject charity requests.
- BR59. The admin shall be able to view system statistics (total donations, number of charities, number of donors, and number of drivers).

- **Driver Requirements**

- BR60. The driver shall be able to sign up using first name, last name, phone number, email, password, profile picture(optional), and city.
- BR61. The driver shall be able to sign in using email and password.
- BR62. The driver shall be able to sign out.
- BR63. The driver shall be able to reset their password using an OTP sent to their email.
- BR64. The driver shall be able to view their profile information that includes first name, last name, phone number, location, email,profile picture and city.
- BR65. The driver shall be able to edit their profile information that includes first name, last name, picture, phone number, email, and city.
- BR66. The driver shall be able to view a list of charities.
- BR67. The driver shall be able to view the charity's profile information, including charity name, license number, profile picture, phone number, brief description, working times and location.
- BR68. The driver shall be able to request to join a charity.
- BR69. The driver shall be able to mark themselves as available or unavailable for deliveries.
- BR70. The driver shall be able to attach their live location when they are available.
- BR71. The driver shall be able to update the status of a delivery (Active, picked up, delivered).
- BR72. The driver shall be able to enter a verification code from the donor to confirm successful pickup.
- BR73. The driver shall navigate to donor and charity locations using Google Maps.
- BR74. The driver shall be able to view all assigned delivery tasks, including donation pickups and drop-off locations, date, and time.
- BR75. The driver shall be able to view their complete delivery history.
- BR76. The driver shall be able to receive an assignment notification for donation post pickup.
- BR78. The driver shall be able to receive assigned delivery tasks when marked as available.

- **System Requirements**

- SR79. The system shall automatically delete food donation posts that remain in an available state for more than 2 days.

6.3 Non-Functional Requirements

6.3.1 Performance

NR1. The system shall respond to user interactions such as login and donation requests within maximum 3 seconds to maintain a smooth user experience.

6.3.2 Security

NR2. The system shall be able to automatically terminate inactive user sessions after 15 minutes of inactivity to prevent unauthorized access from unattended devices.

NR3. The system shall sanitize all fields on the register, ensuring that each input is limited to 300 characters.

NR4. The system shall be able to validate and restrict all image uploads to PNG or JPEG formats with a maximum size of 3 MB to prevent malicious files from being stored.

NR5. The User passwords shall have a minimum length of 8 characters.

6.3.3 Availability

NR6. Maintenance and updates shall cause within maximum time 10 minutes of downtime per week.

6.3.4 Interoperability

NR 7. The system shall integrate with Google Maps.

6.3.5 Reliability

NR 8. The system shall automatically recover from minor within maximum time 10 seconds without data loss.

6.3.6 Ease of use

NR 9. The new user shall be able to learn how to use the system within maximum 10-20 minutes.

NR 10. After training, users shall be able to perform any task with a maximum of 5 minutes.

6.3.7 Legal & Ethical Compliance

NR 11. The system shall comply with Saudi data protection and charity regulations.

6.4 Design Constraint

DC1. The system shall support the English language.

DC2. The system shall be available in 10 months.

DC3. The system shall support Android mobile applications for donors,charities and drivers.

DC4. The system shall be able to integrate with Google Maps API.

DC5: The system shall be developed as a web based application for admin.

DC6: The system's data shall be stored in the Cloud Firestore database.

DC7: The system;s database shall be a real-time database.

7. Problem Complexity

The **FAYDH** donation management system presents a high level of complexity due to the wide range of technical, organizational, and social factors involved in its development and operation. Several aspects contribute to this complexity:

1. **Multi-User Environment**

The system serves diverse user categories donors, charities and drivers, each requiring distinct interfaces, workflows, and permissions. Managing interactions and data flow between these user groups increases both design and implementation complexity.

2. **Integration with External Services**

The system integrates with third-party systems, including Google Maps APIs for location and navigation. Ensuring smooth interoperability, compatibility, and reliability across these external components presents a significant technical challenge.

3. **Real-Time Tracking and Notifications**

The system must handle **real-time updates** for donation status, pickup scheduling, and delivery verification using **verification codes**. Synchronizing these events across all users without data loss or delay requires efficient backend design and real-time database handling.

4. **Data Security and Privacy Concerns**

Protecting sensitive user data, including identity documents and personal information of donors, charities and drivers is critical. Implementing strong encryption, secure authentication, and privacy-preserving mechanisms introduces complexity in both the architectural and operational layers.

5. **Regulatory and Ethical Compliance**

Since the project operates within Saudi Arabia's charitable sector, it must comply with local charity regulations, data protection policies, and safety standards for various types of donations, including food, clothing, furniture, and flowers. Ensuring that the technical implementation adheres to these legal and ethical frameworks while maintaining humanitarian integrity adds a layer of complexity that goes beyond purely technical challenges. This includes verifying that donated items are safe, hygienic, and suitable for use or consumption, protecting donors', charities', and drivers' data, and respecting local laws and organizational policies for all types of donations.

6. **Scalability and Performance Management**

The system is expected to handle **hundreds of concurrent users** performing various operations, ensuring consistent performance, minimal downtime, and high availability requires careful system optimization and infrastructure design.

7. **Cross-Disciplinary Nature**

The project involves multiple disciplines—software engineering, data security, UX design, and humanitarian domain knowledge—making coordination and integration among these areas a complex task.

8. **The smart matching algorithm**

The system must efficiently assign donations to the nearest available drivers while accounting for urgent items like food or flowers, ensuring accurate real-time distance calculations, minimal assignment delays, and scalable performance under potentially high volumes of concurrent donation requests.

8. System Use Cases

The use case diagram illustrates the functional interactions between users Donor, Charity, Driver, and Admin and the **FAYDH** system.

8.1 Use Case Diagram

8.1.1 Donor Use Case Diagram



Figure 8. Donor use case diagram

8.1.2 Charity Use Case Diagram

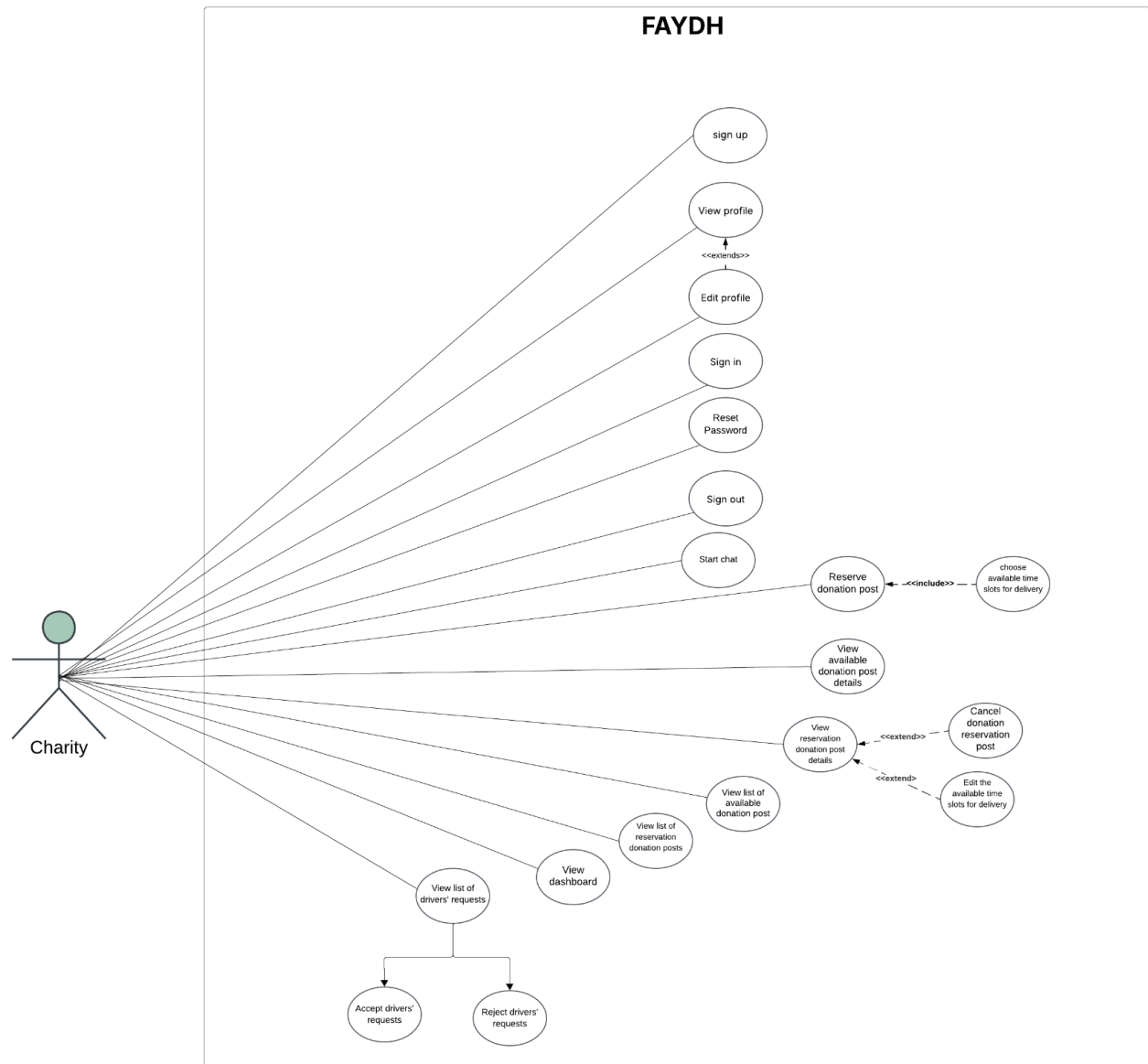


Figure 9. Charity use case diagram

8.1.3 Driver Use Case Diagram



Figure 10. Driver use case diagram

8.1.4 Administrator Use Case Diagram

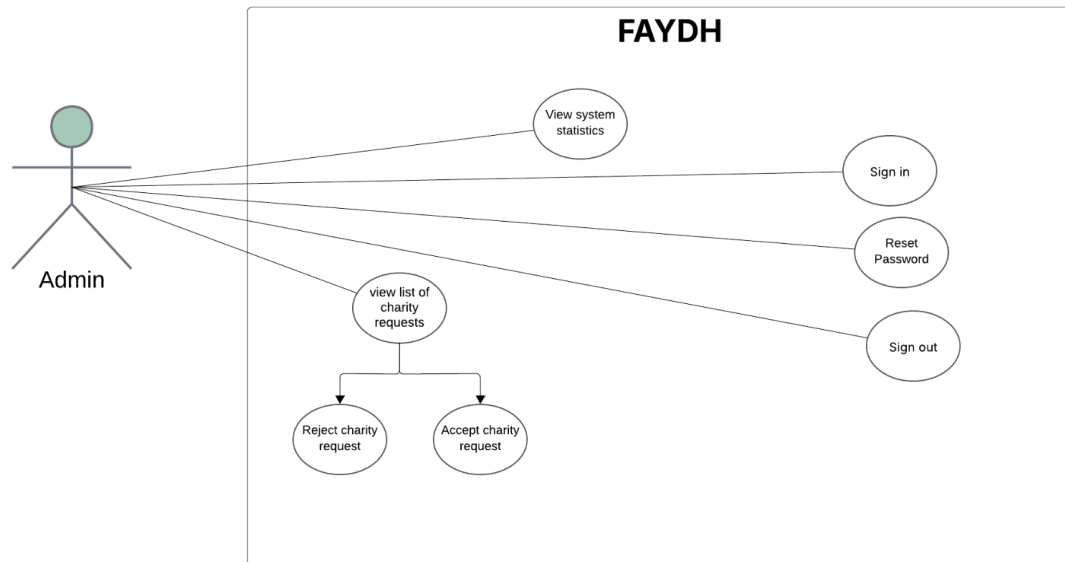


Figure 11. Admin use case diagram

8.1.5 Full Use case

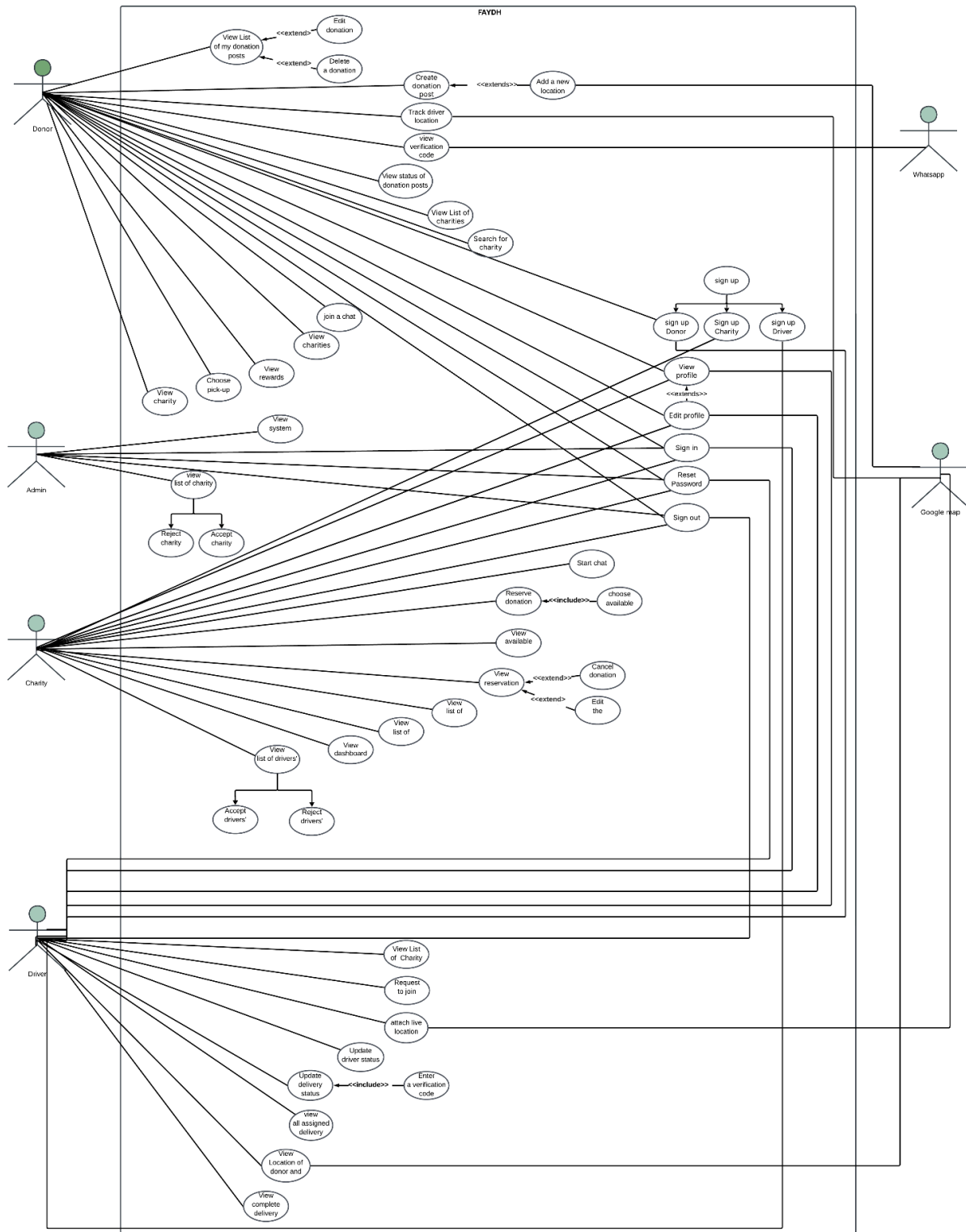


Figure 12. Full use case diagram

8.2 Use Case Descriptions

This section provides a detailed and structured explanation of how specific system functions operate from the perspective of both the actors and the system. It describes the interactions between the primary actor, secondary actors, and the system, while clarifying the relationships, preconditions, and postconditions that govern the execution of each scenario.

8.2.1 Create Donation Post Use Case description

Use Case Descriptions		
System: <u>FAYDH</u> Application		
Use Case Name: Create Donation Post		
Primary Actor: Donor	Secondary Actor(s): Google map,charity	
Description: This case describes how a donor creates a new donation post.		
Relationships: Includes: None. Extends: Add a new location. Generalization: None.		
Pre-conditions: 1- The donor is signed in into the system.		
Steps:		
Primary Actor (Donor)	System	Secondary actor(s)
1. The donor selects “New donation”. 3. The donor selects a donation type. 5.The donor fills in the required donation post form. 7. The donor selects an existing location. 8.The donor submits the donation post. 10. The donor confirms the message. 13.The donor views the successful message.	2. The system displays available donation types (Food, Clothes, Flowers, Furniture). 4.The system displays the relevant donation form. 6.The system validates the input. 9.The system displays confirmation messages. 11. The system saves the donation post. 12.The system displays a successful message.	

	14. The system sends a notification to all charities indicating that the new donation post has been available.	
Alternative and Exceptional Flows: In step 6: if the donor does not have an existing location or wants to use a new one: The donor executes the “Add a new location” use case. In step 7: If required fields are empty or contain invalid data: The system highlights errors and displays an appropriate error message (e.g., “Photo required”, “Invalid quantity”).		
Post-conditions: Successful Condition: The donation post is stored in the database, and the post appears in the donor’s “My Donation Posts” list with status available and a notification sent to charities. Failure Condition: The donation post is not created due to an error (missing data or system failure), an error message displayed (the donation post cannot be created due to system issues).		

Table 3.Create Donation Post use case description

8.2.2 Choose Pickup Available Time Use Case Description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: Choose Pickup Available Time		
Primary Actor: Donor	Secondary Actor(s): None	
Description: This use case allows the donor to choose a specific time slot from charity for when their donation can be picked up.		
Relationships: Includes: None Extends: None Generalization: None		
Pre-conditions: 1- The donor is signed into the system. 2- The charity has already reserved a donation post.		
Steps:		
Primary Actor (donor)	System	Secondary actor(s)
1-The donor selects a donation post already reserved by charity. 3-The donor selects a preferred pickup date and time. 4-The donor submits a pickup date and time. 6-The donor confirms the selected pickup date and time. 10- The donor views the successful message.	2-The system displays available pickup time slots. 5- The system displays a confirmation message. 7-The system stores the chosen time. 8-The system updates the donation post with the scheduled pickup. 9-The system displays a successful message.	
Alternative and Exceptional Flows:		
In any step before confirmation(step 6): if the Donor Cancels the Process		
1. The system does not save any changes. 2. The system returns to the donation post or previous page.		
Successful Condition: The donor’s selected pickup time is successfully saved in the system, and the donation post is updated with the chosen pickup schedule.		
Failure Condition: The selected time is not stored due to a system error or internet issue.		

Table 4.Choose Pickup Available Timet use case description

8.2.3 Join chat Use Case Description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: Join chat		
Primary Actor: Donor	Secondary Actor(s): None	
Description: This Use case describes how a donor can join a chat with the charity after the post is reserved by charity.		
Relationships: Includes: None Extends: None. Generalization: None.		
Pre-conditions: 1-The donor is signed in into the system. 2-The charity executes the “Start chat” use case for reserved posts. 3-The donor receive the notification about new chat session.		
Steps:		
Primary Actor (Donor)	System	Secondary actor(s)
1- The donor selects a chat tab. 3- The donor selects a Specific chat. 6- The donor views the chat screen.	2-The system retrieve and displays all chat session 4-The system retrieve and validate Donation PostStatus for selected chat if it still reserved or turn to delivered. 5- The system displays the chat screen in real time and synchronizes subsequent messages between both parties Charity and Donor.	
Alternative and Exceptional Flows: In step 4: if the status of the donation post is already turn to delivered: 1. The system closes the chat and displays : “this chat is no longer available”. 2. The donor can't be able to send new messages to charity.		
Post-conditions: Successful Condition: The donor can join a chat successfully; both charity and donor can exchange messages, Failure Condition: The donor could not be join to a chat due to an invalid chat room or system error. If the system fails to open the chat due to errors (network, permission, or database issue), it displays an error message: “Unable to open chat. Please try again later.”		

Table 5.Join chat use case description

8.2.4 Reserve donation post Use case description

Use Case Description		
System: FAYDH Application		
Use Case Name: Reserve donation post		
Primary Actor: Charity	Secondary Actor(s): None	
Description: This Use case describes how a charity reserves available donation posts from donors.		
Relationships: Includes: Select the available time slots for delivery. Extends: None. Generalization: None.		
Pre-conditions: 1-The charity is signed in into the system. 2- The charity executes “View donation post details” use cases.		
Steps:		
Primary Actor (Charity)	System	Secondary actor(s)
1-The charity selects “Reserve post”. 3-The charity executes “choose available time slots for delivery” use case. 5-The charity confirms the reservation. 9- The charity views a successful message.	2- The system validates the donation post status. 4-The system displays a confirmation message for post reservation. 6- The system updates the post status to “Reserved”. 7-. The system removes the post from other charities. 8- The system displays a successful reservation message. 10- The system sends a reservation notification to the donor.	
Alternative and Exceptional Flows:		
In step 2: if the charity attempts to reserve a post that is reserved by another charity.		
1.The system displays a message: “This donation post has already been reserved by another charity.” 2.The reservation process is terminated.		
In step 2: if the charity attempts to reserve a post that is expired.		

1.The system displays a message: "This donation post has expired". 2.The reservation process is terminated. In step 2: if the charity attempts to reserve a post that is deleted. 1.The system displays a message: "This donation post has been deleted". 2.The reservation process is terminated. In step 4: if the charity cancels the reservation before confirmation. 1.The reservation process ends without saving any changes.
Post-conditions: Successful Condition: The post status is updated to"Reserved",it is no longer available to other charities,the selected date and time are published successfully and the donor is notified about the reservation. Failure Condition: If the system fails to process reservation requests, it displays an error message (Reservation cannot be completed due to system issues). And the post status remains "Available".

Table 6. Reserve donor post use case description

8.2.5 Cancel donation reservation post Use case description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: Cancel donation reservation post		
Primary Actor: Charity	Secondary Actor(s):None	
Description: This Use case describes how a charity can cancel a previously reserved donation post from a donor before delivery.		
Relationships: Includes: None. Extends: None. Generalization: None.		
Pre-conditions: 1-The charity is signed in into the system. 3- The charity executes the " View reservation donation post details " use case.		
Steps:		
Primary Actor (Charity)	System	Secondary actor(s)
1-The charity selects “Cancel Reservation”. 3-The charity confirms the cancellation. 7- The charity views a success message.	2-The system displays a confirmation message. 4-The system updates the donation post status to “Available.” 5-The system removes the post from the “My Reservations" page. 6- The system displays a successful cancellation message. 8- The system sends a notification to the donor indicating that the reservation has been canceled.	
Alternative and Exceptional Flows: In step 2: if the system fails to retrieve the donor post details: 1. The system displays a message: “Unable to load donation post details. Please try again later.”		
Post-conditions: Successful Condition: The donation post is updated to “Available,” and the donor is notified that the charity has canceled the reservation. Other charities can now view and reserve the post again. Failure Condition: If the system fails to cancel the reservation due to a system error (network or database issue), it displays an error message (Cancellation cannot be completed due to system issues), and the post status remains “reserved”		

Table 7.Cancel donation reservation post use case description

8.2.6 View available donation post details Use Case description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: View available donation post details		
Primary Actor: Charity	Secondary Actor(s): None	
Description: This Use case describes how a charity can view available donation post details including donor name, phone number, email, and donation form relevant to donation type.		
Relationships: Includes: None. Extends: None. Generalization: None.		
Pre-conditions: 1-The charity is signed in into the system. 2- The charity executes a “View list of available donation post” use case.		
Steps:		
Primary Actor (Charity)	System	Secondary actor(s)
1- The charity selects a specific donation post to view.	2- The system retrieves and displays detailed information about the selected post, including: donor information (name, phone number, email) , donation type, donation form relevant to donation type.	
Alternative and Exceptional Flows:		
In step 2: If the donor deleted the post when the charity views it: 1. The system displays a message (This donation post is no longer available). 2. The system redirects the charity to the donation posts list.		
In step 2: If the post status is turn to “reserved” when the charity views it: 1. The system displays a message (This donation post is no longer available). 2. The system redirects the charity to the donation posts list.		
In step 2: If the post expired when the charity views it: 1. The system displays a message (This donation post is no longer available). 2. The system redirects the charity to the donation posts list.		
Post-conditions:		
Successful Condition: The charity successfully views all details of the donation post including donor name, phone number, email, donation type and donation form.		
Failure Condition: If the post is unavailable due to a system error (network or database issue), an error message is displayed (post details cannot be viewed due to system issues).		

Table 8.View available donation post details use case description

8.2.7 Choose Available Time Slots for Delivery Use Case Description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: Choose Available Time Slots for Delivery.		
Primary Actor: Charity	Secondary Actor(s): None	
Description: This use case describes how the charity chooses available dates and time slots for delivery to the donor after reserving a donation post.		
Relationships: Includes: None. Extends: None. Generalization: None.		
Pre-conditions: 1- The charity is signed in. 2- The charity is in the reservation process.		
Steps:		
Primary Actor (Charity)	System	Secondary actor(s)
2- The charity chooses suitable dates and time ranges for delivery.	1- The system displays the dates and time ranges.	
Alternative and Exceptional Flows:		
In step 2: Overlapping Time Slots		
1. The charity chooses date and time slots that overlap with previously published slots.		
2. The system detects the invalid selection.		
3. The system displays an error message prompting the charity to adjust the time slots.		
Post-conditions:		
Successful Condition: The time slots are successfully selected.		
Failure Condition: If the selection failed due to a system error (network or database issue), an error message displayed (selection failed due to system issues).		

Table 9.Choose Available Time Slots for Delivery use case description

8.2.8 Request to join charity Use Case Description

Use Case Descriptions		
System: <u>FAYDH</u> Application.		
Use Case Name: Request to join charity		
Primary Actor: Driver	Secondary Actor(s): charity	
Description: This use case describes how a driver requests to join a charity.		
Relationships: Includes: None. Extends: None. Generalization: None.		
Pre-conditions: 1- The driver is signed in into the system. 2- The driver already views a list of charity.		
Steps:		
Primary Actor (Driver)	System	Secondary actor(charity)
1.The driver selects a charity to view the charity profile. 3.The driver requests to join a charity. 5-The driver confirms the message. 8-The driver views a successful message.	2.The system displays the charity profile. 4. The system displays a confirmation message. 6- The system stores the driver's request to the database 7-The system displays a successful message. 9-The system sends a notification to the requested charity.	
Alternative and Exceptional Flows: In Step 3: if the driver already has a pending request for the same charity: The system displays a message: "Request already pending. Please wait for approval." In Step 3: if the driver's request has already been declined by the charity: The system displays a message: "Request has been rejected by charity."		
Post-conditions: Successful Condition: The join request is saved in the database and is sent to the charity with pending status. Failure Condition: The join request is not sent due to an error (missing data or system failure), an error message displayed (the request cannot be created due to system issues).		

Table 10.Request to join charity use case description

8.2.9 Update delivery status Use Case description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: Update delivery status		
Primary Actor: driver	Secondary Actor(s): None	
Description: This use case describes how a driver updates the status of an assigned delivery.		
Relationships: Includes: Enter a verification code. Extends: None. Generalization: None.		
Pre-conditions: 1- The driver is signed into the system. 2- The driver selects a delivery assignment.		
Steps:		
Primary Actor (driver)	System	Secondary actor(s)
2- The driver views the details. 3- When the driver arrives at the donor location to pick up, execute the “Enter a verification code” use case from the donor. 5- When the driver arrives at the charity drop-off, the driver selects “Delivered”.	1- The system displays the active delivery details including the donor name and phone number, the pick-up (donor location) and drop-off locations. 4- The system updates delivery status to “Picked Up”. 6-. The system updates the status from Picked Up to Delivered.	
Alternative and Exceptional Flows: In step 5, The driver taps “Delivered” without previously completing the pick-up step. 1. The system displays a warning: “You must pick up the delivery before marking it as Delivered”. 2. The driver is redirected to the pick-up verification step.		
Post-conditions: Successful Condition: The system successfully updates the delivery status from Active → Picked Up → Delivered. Failure Condition: The delivery status is not updated due to an error (missing data or system failure), an error message displayed (the delivery status can not be updated due to system issues).		

Table 11.Update delivery status use case description

8.2.10 Sign Up charity Use Case Description

Use Case Descriptions		
System: FAYDH Application		
Use Case Name: Sign up Charity		
Primary Actor:, Charity	Secondary Actor(s): None	
Description: This use case describes how new Charities create an account in the system.		
Relationships: Includes: None. Extends: None. Generalization: None.		
Pre-conditions: 1- The charity is not already registered in the system. 2- The charity has already selected his Sign up role.		
Steps:		
Primary Actor (Charity)	System	Secondary actor(s)
2- The charity fills in the required information: charity name, license number, Phone number, Email, password, profile picture (optional), Brief Description, and Location. 4- The charity selects Sign Up.	1- The system displays the Sign up form. 3- The system validates the format of required entered data. 5- The system stores the account in the database as a “pending” account.	
Alternative and Exceptional Flows		
In step 3 : missing input The system displays an error message and asks the user to fill the empty field.		
In step 3: Email, License Number, or Phone Number already exists The system notifies the user that the email is already in use and prompts them to use another email or sign in.		
In step 3: invalid format for required input The system displays an error message and asks the user to refill the fields with the correct format specified.		
Post-conditions: Successful Condition: The charity account is created and stored in the system database as “pending”. Failure Condition: The account is not created due to an error (missing data or system failure), an error message displayed (the charity is not created due to system issues).		

Table 12.Sign up charity use case description

9. Analysis Class

This section contains the analysis class diagrams which includes all necessary boundary, control, and entity classes of important use-cases and their relationships.

9.1 Create donation post

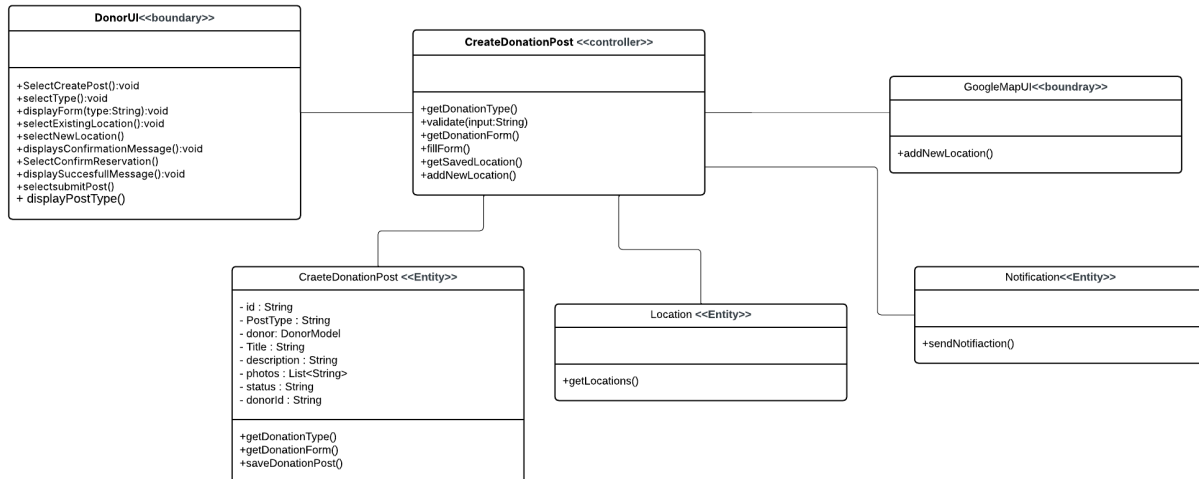


Figure 13. Create donation post VOPC

9.2 Choose pickup available time

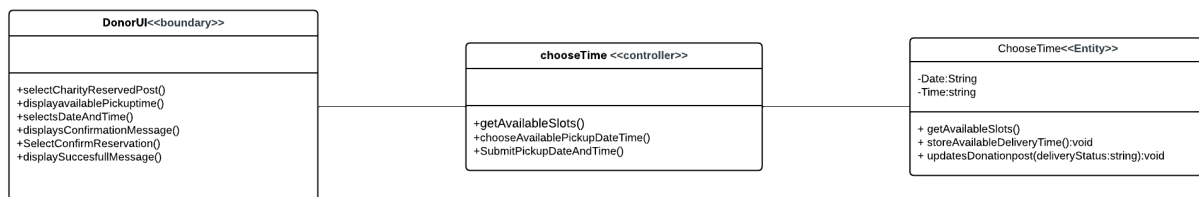


Figure 14. Choose pickup available time VOPC

9.3 Join chat

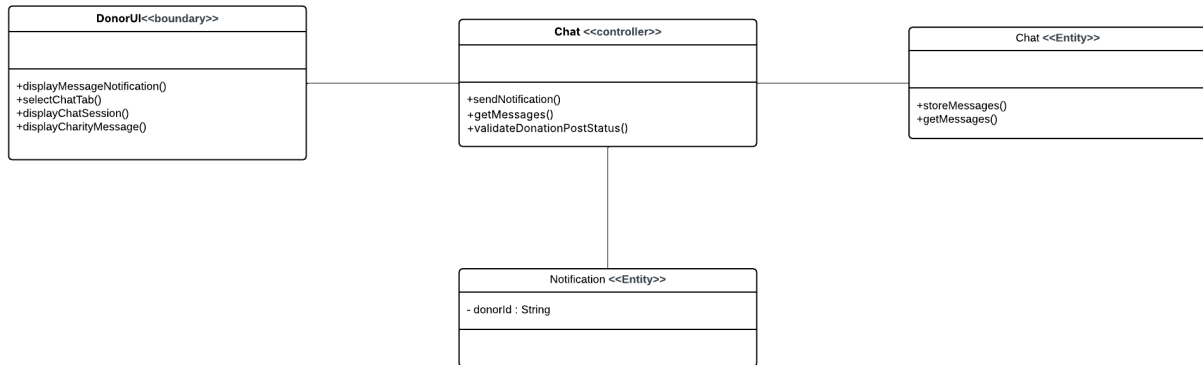


Figure 15. Join chat VOPC

9.4 Reserve donation post

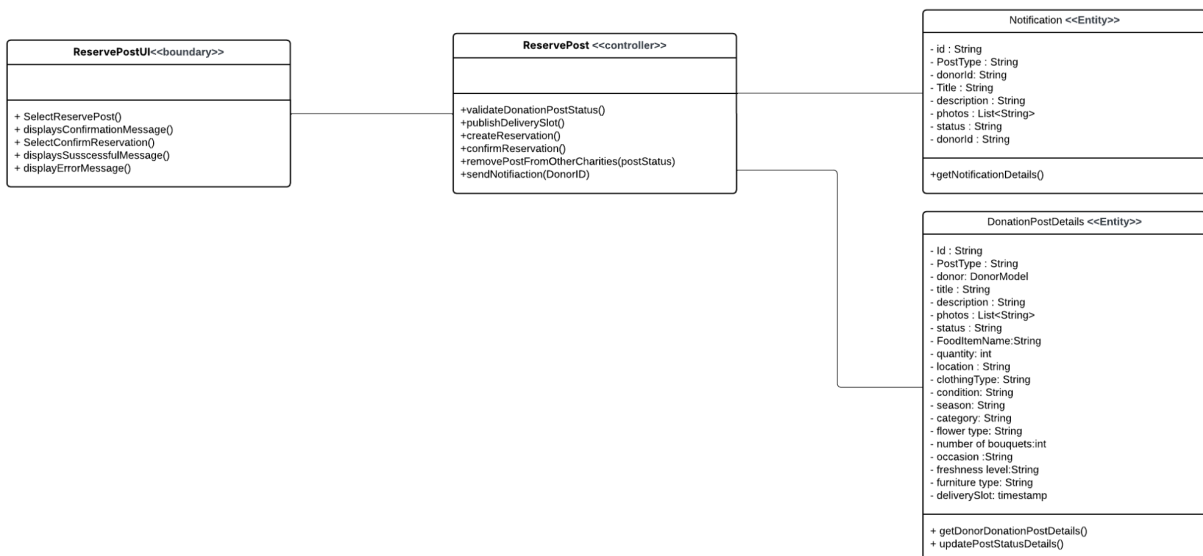


Figure 16. Reserve donation VOPC

9.5 Cancel donation reservation post

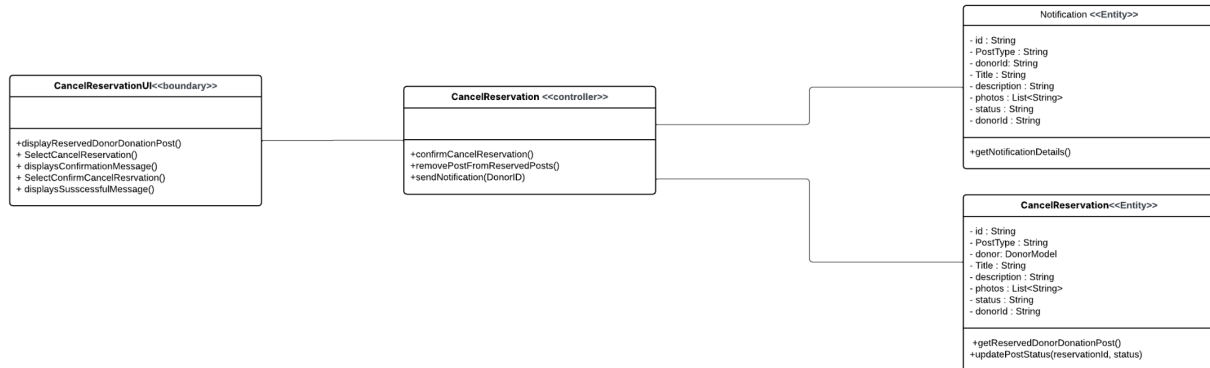


Figure 17. Cancel donation reservation VOPC

9.6 View available donation post details

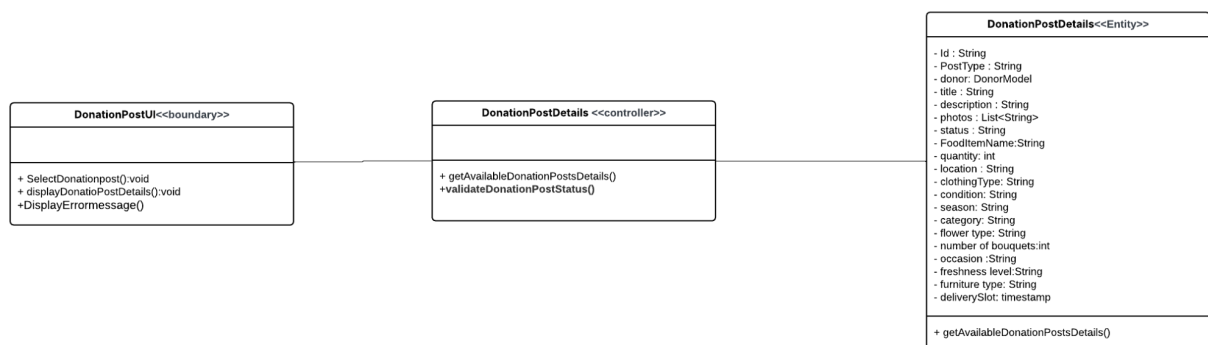


Figure 18. View available donation post details VOPC

9.7 Choose available time slots for delivery

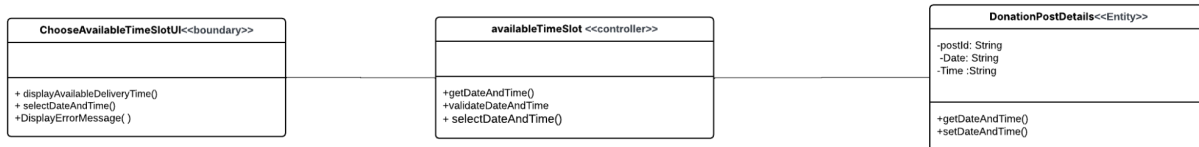


Figure 19. Choose available time slots for delivery VOPC

9.8 Request to join charity

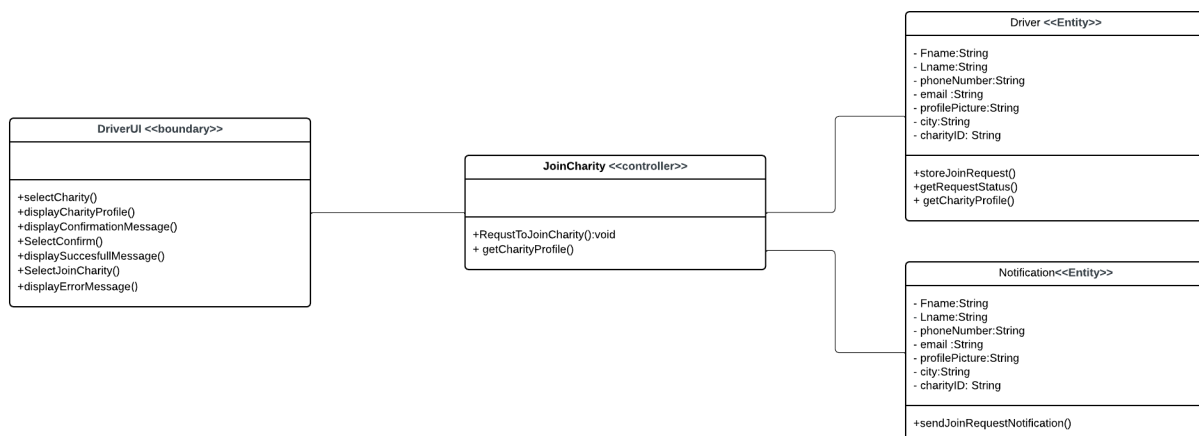


Figure 20. Request to join charity VOPC

9.9 Update delivery status

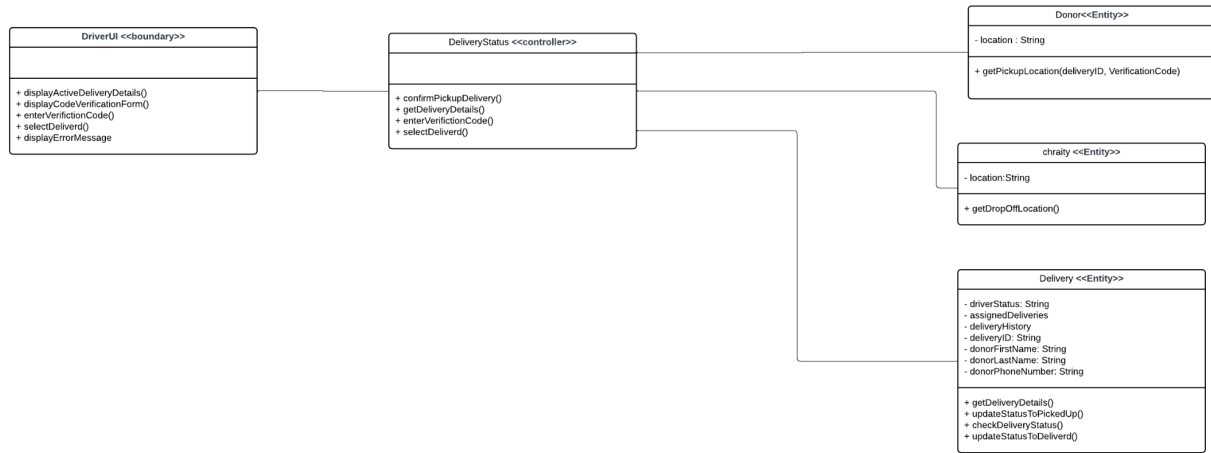


Figure 21. Update delivery status VOPC

9.10 Sign up charity

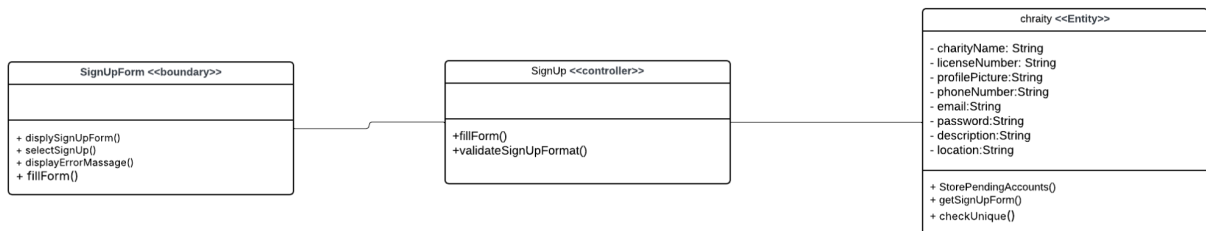


Figure 22. Sign up charity VOPC

10. Interaction Diagram

This section contains the interaction diagrams of the system. The interaction diagram shows the basic flow between the system's actors and objects using sequence diagrams.

10.1 Create Donation Post

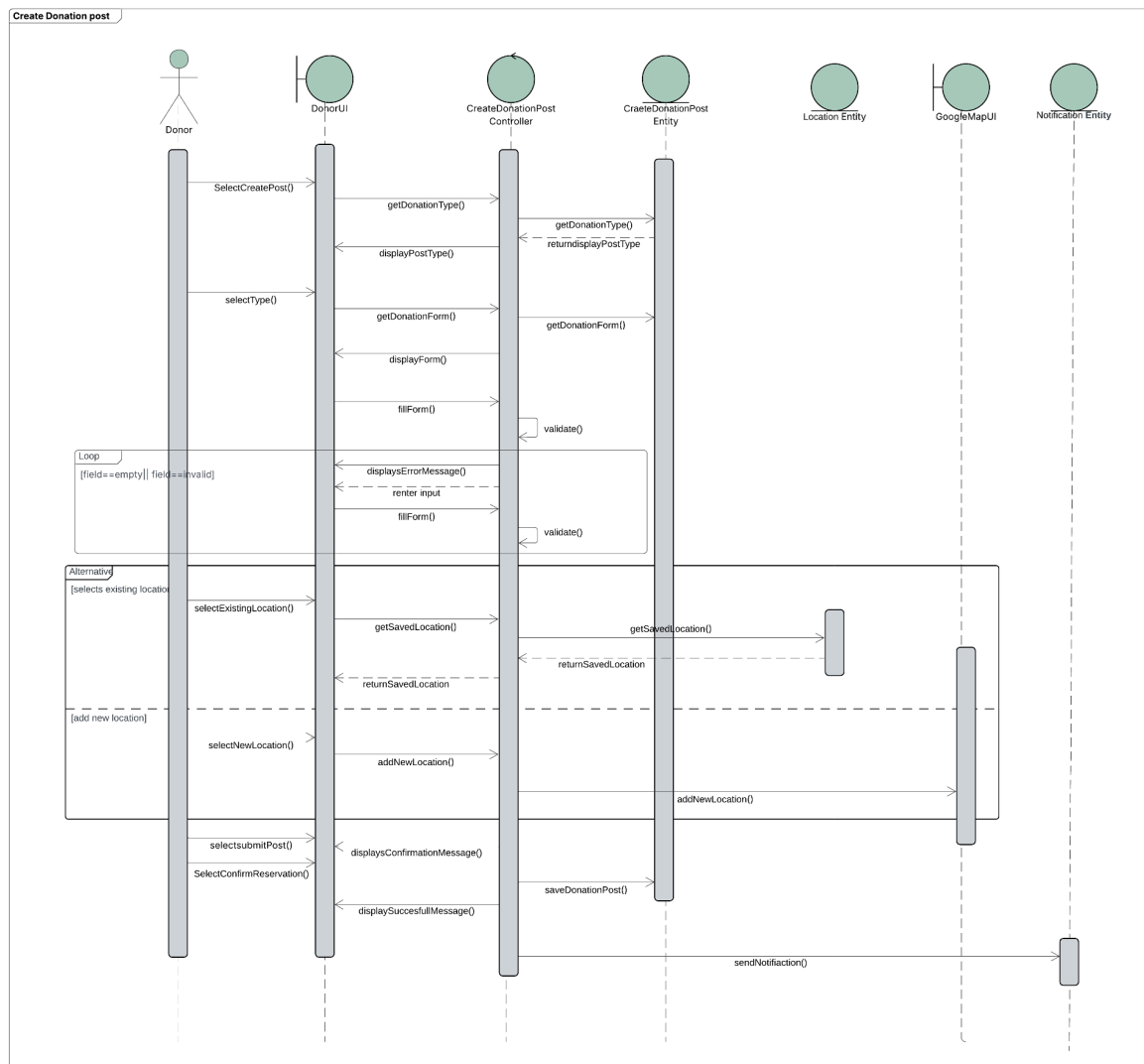


Figure 24. Create donation post sequence diagram

10.2 Choose pickup available time

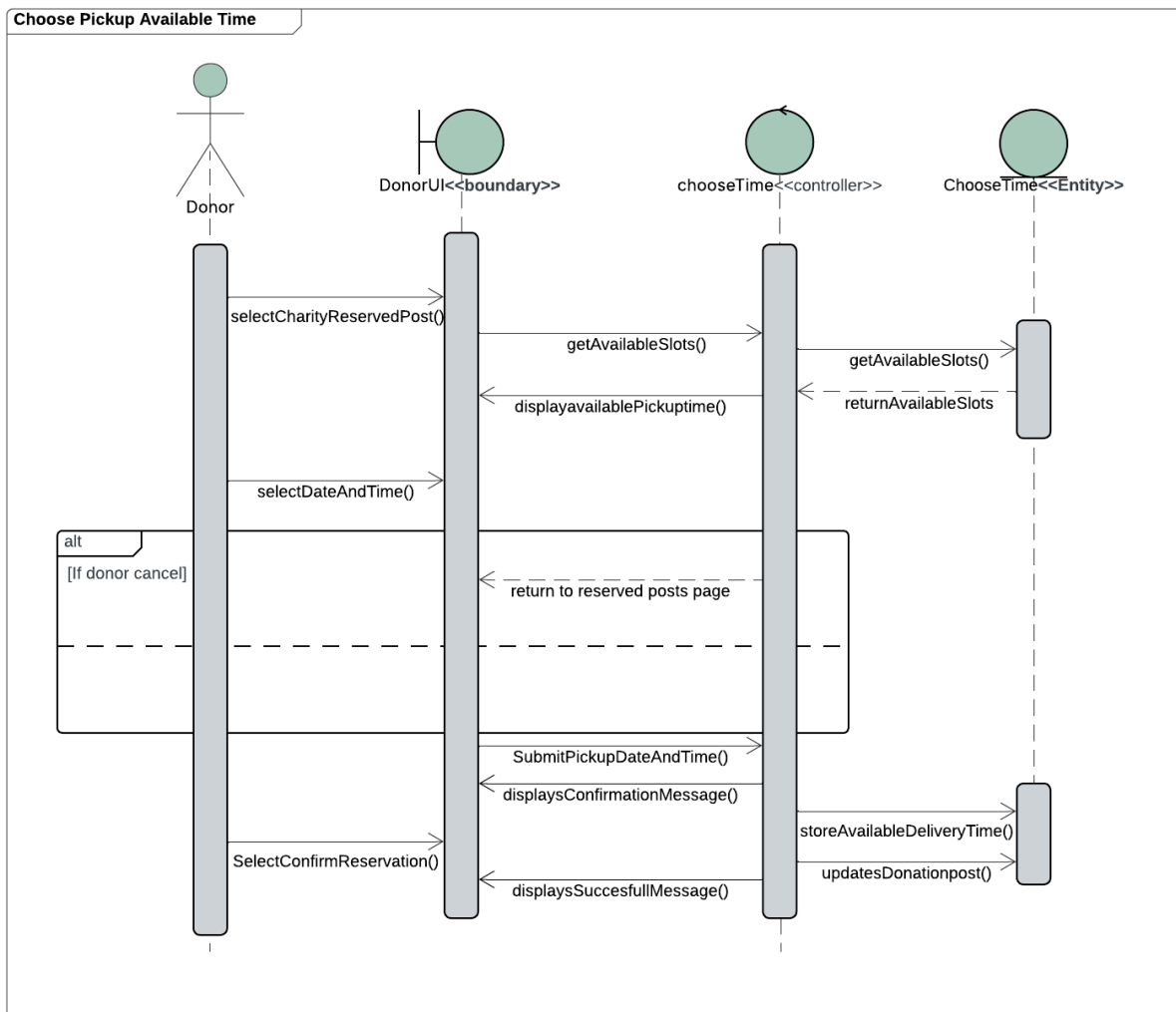


Figure 25. Choose pickup available time sequences diagram

10.3 Join chat

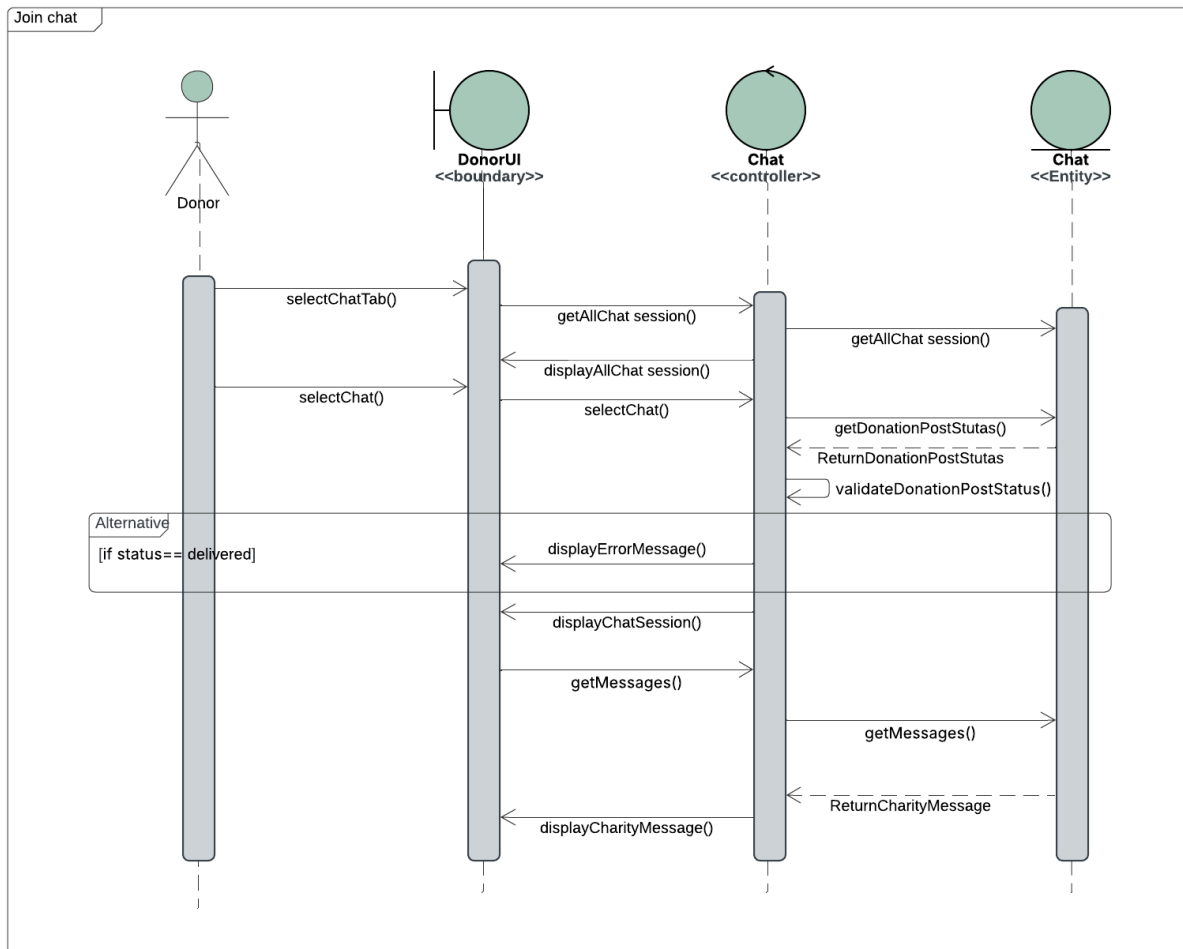


Figure 26. Join chat sequences diagram

10.4 Reserve donation post

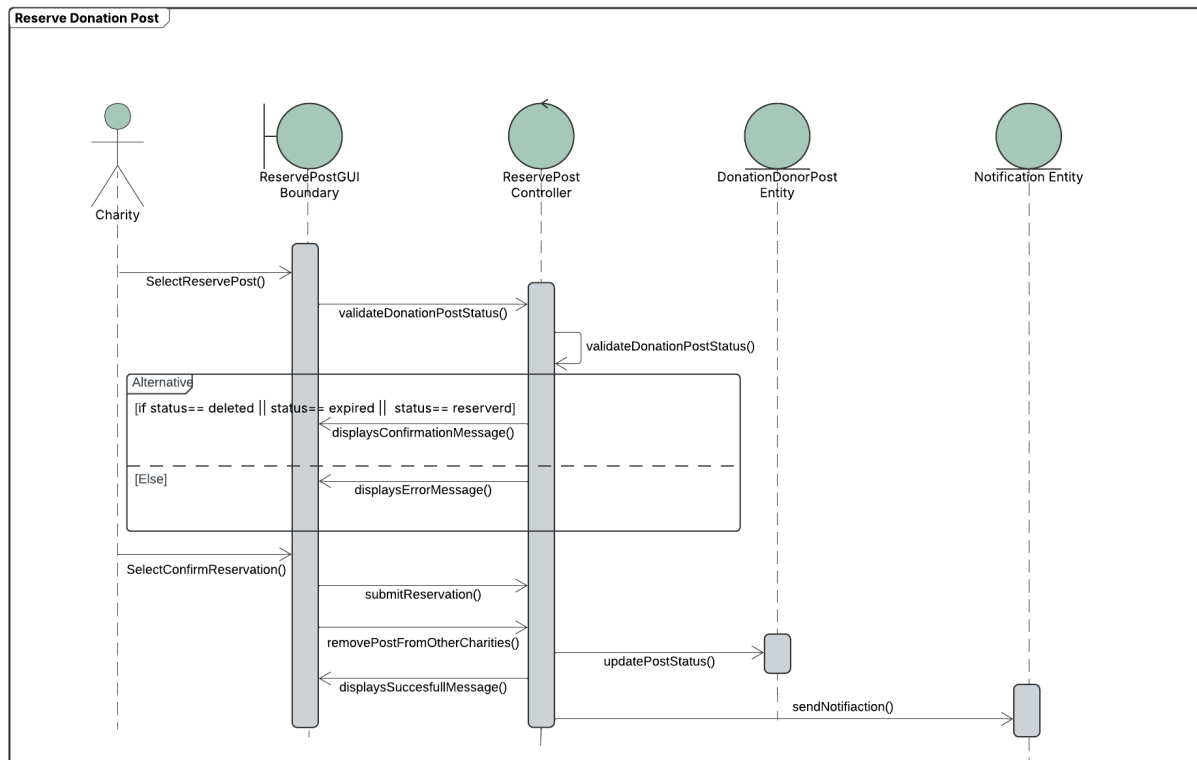


Figure 27. Reserve donation post sequences diagram

10.5 Cancel donation reservation post

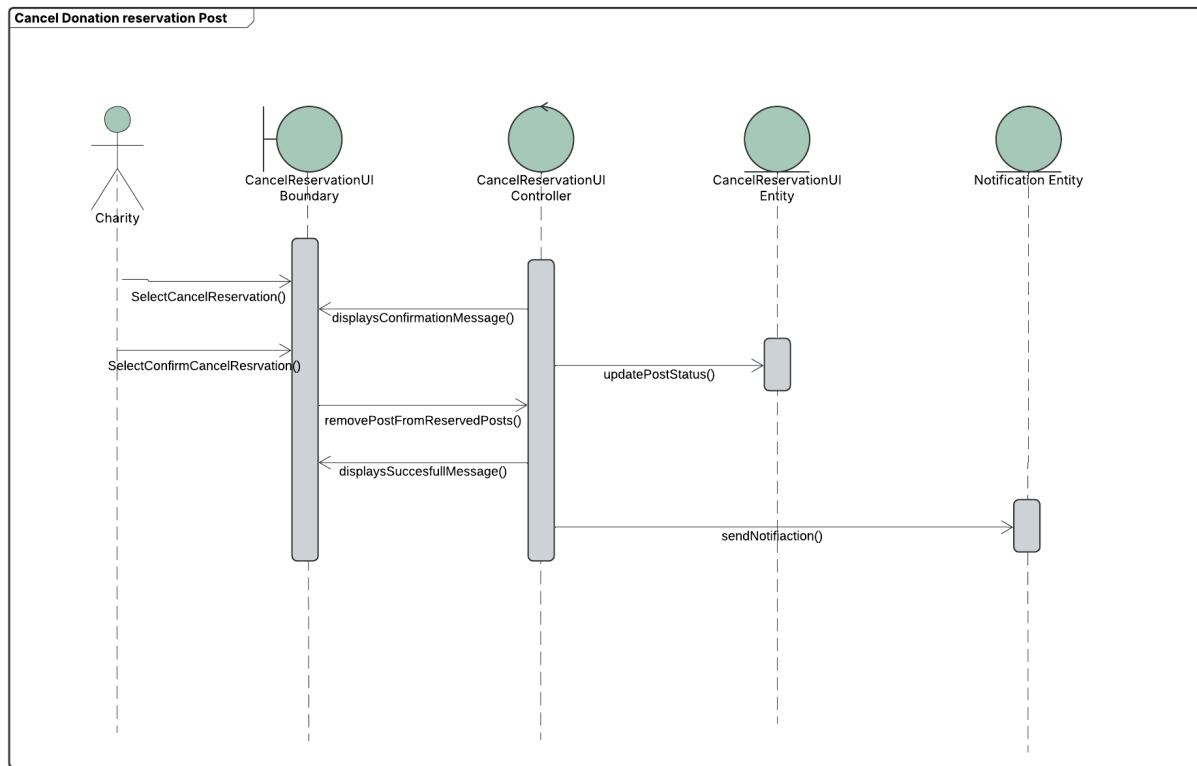


Figure 28. Cancel donation reservation post sequences diagram

10.6 View available donation post details

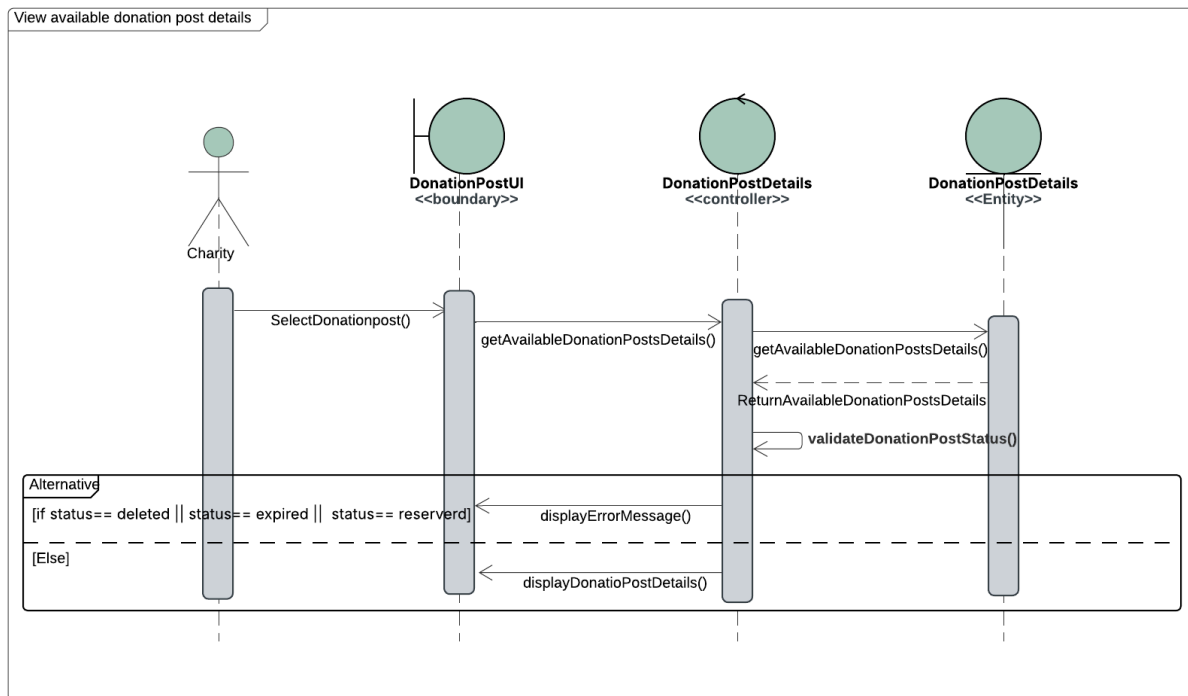


Figure 29. View available donation post details sequences diagram

10.7 Choose available time slots for delivery

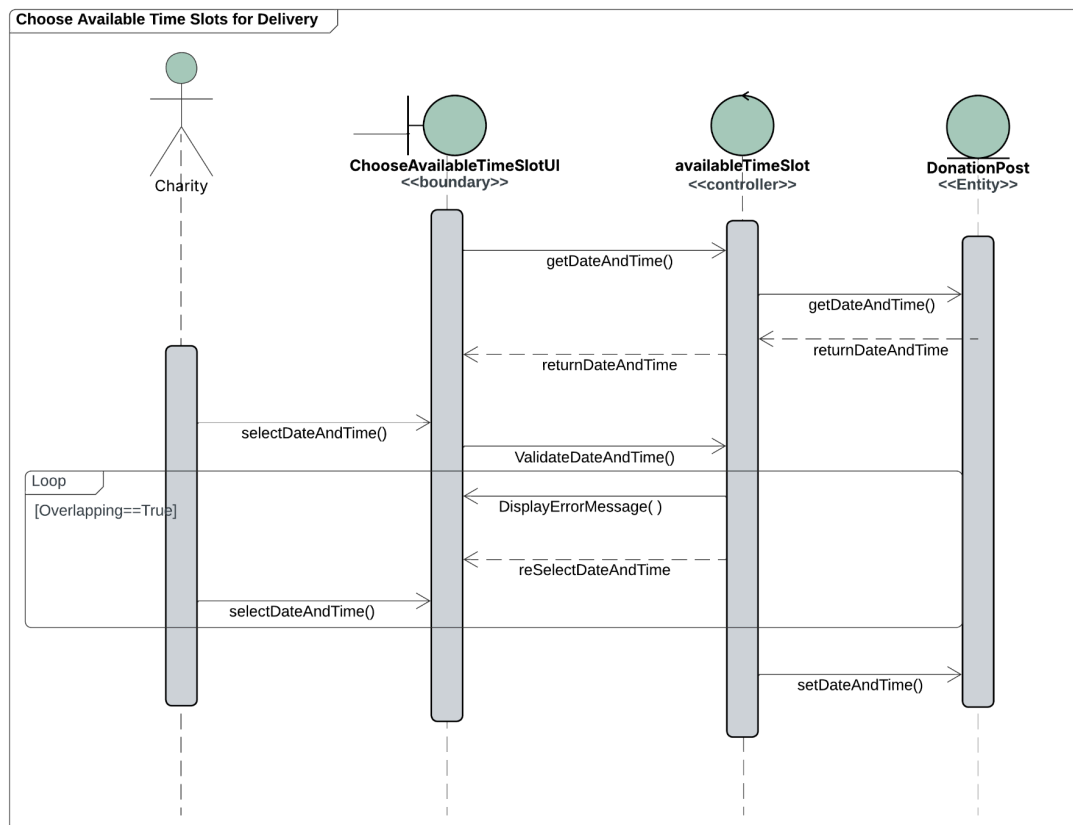


Figure 30. Choose available time slots for delivery sequences diagram

10.8 Request to join charity

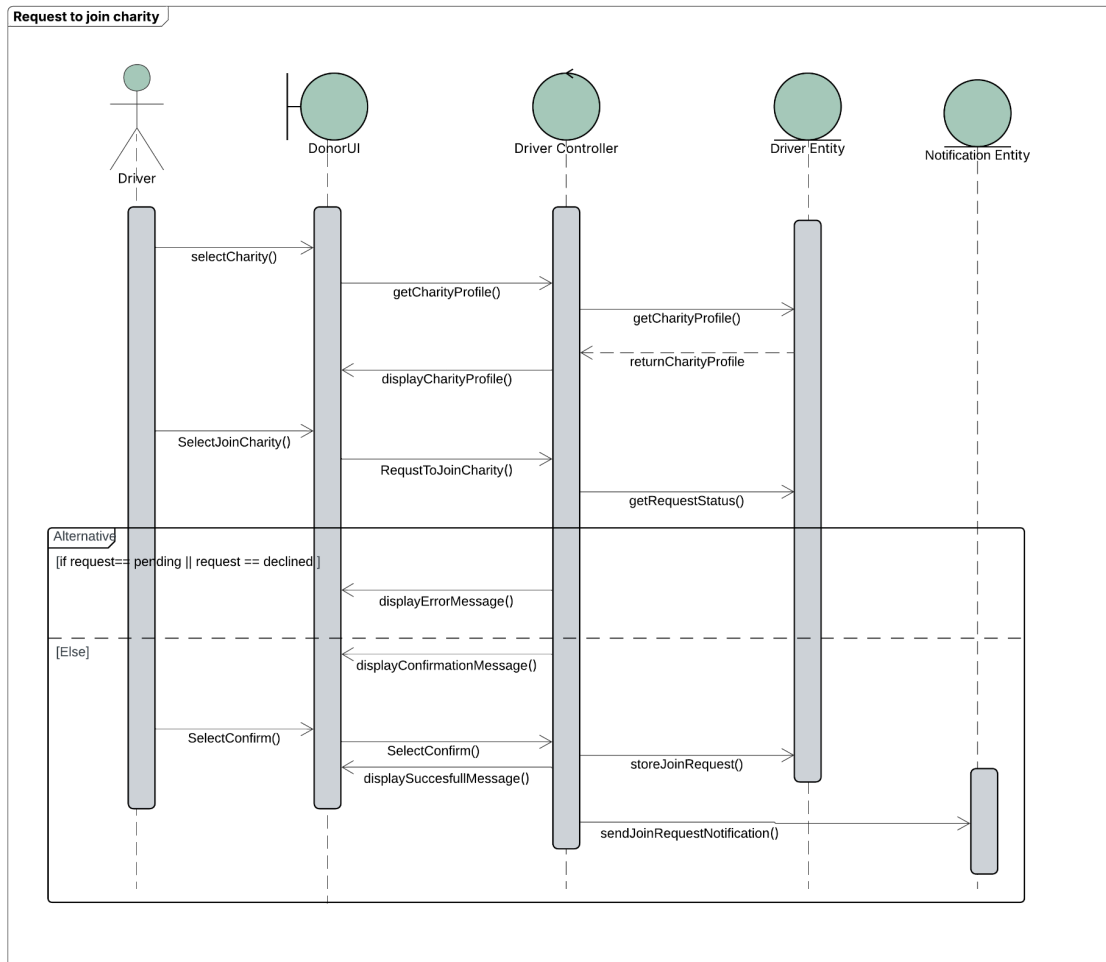


Figure 31. Request to join charity sequences diagram

10.9 Update delivery status

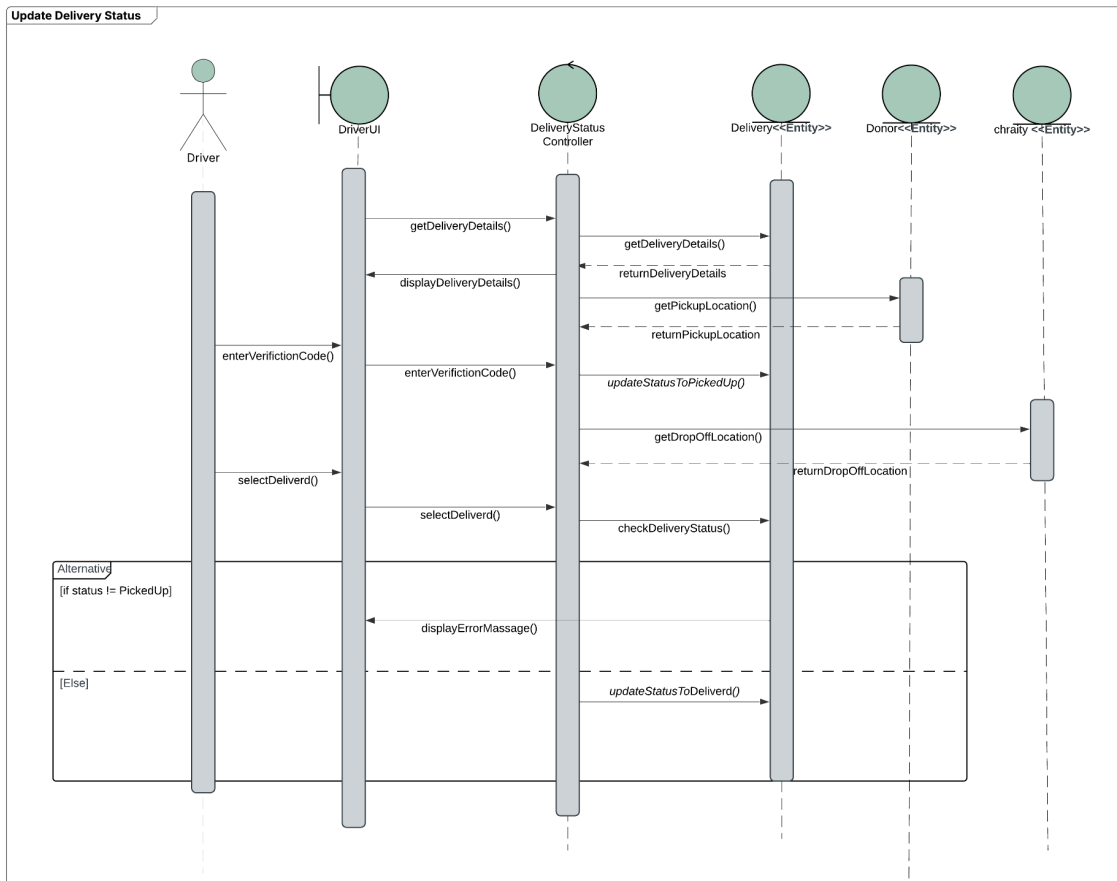


Figure 32. Update delivery status sequences diagram

10.10 Sign Up charity

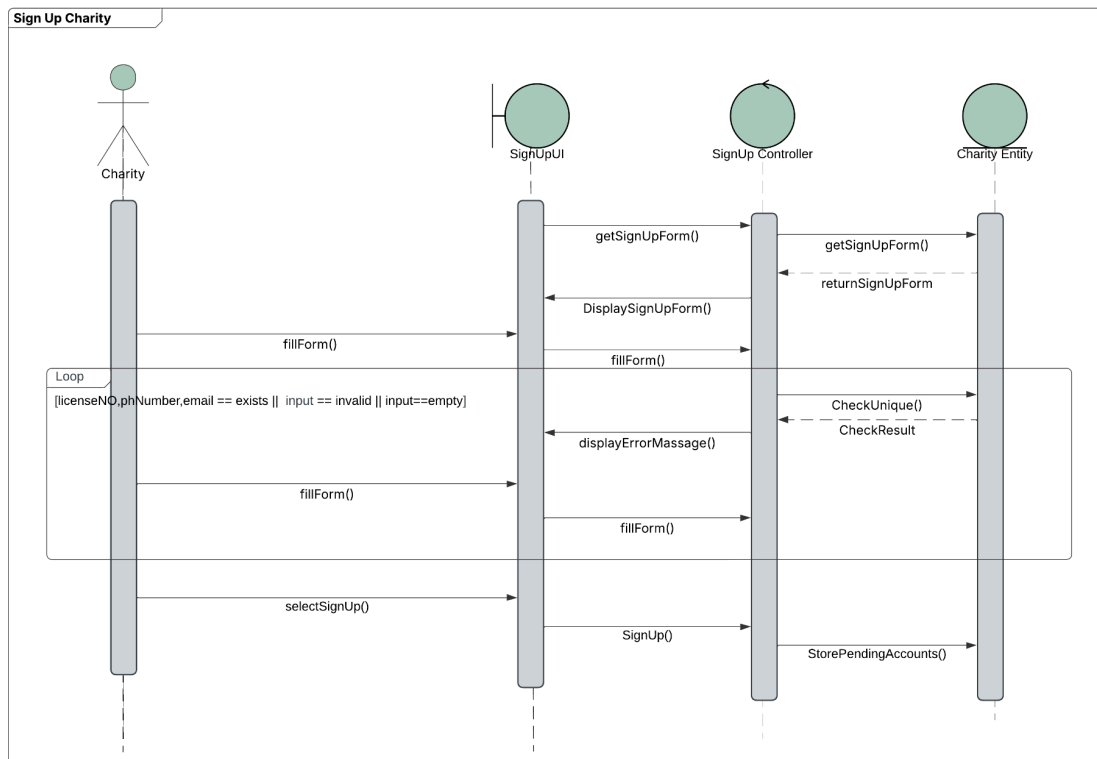


Figure 33. Sign up charity sequences diagram

All the Interaction diagrams above were illustrated in Google Drive [].

11. Design Class

This section shows the design class for the **FAYDH** system. Each class's attributes and methods are displayed, along with the relationships and associations between them. Use this link for a clearer view.

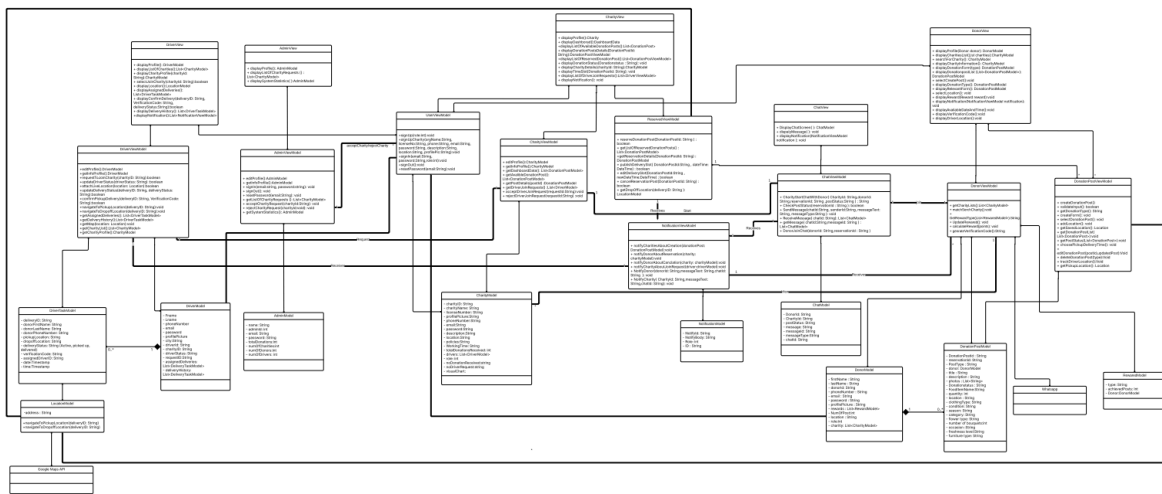


Figure 34. **FAYDH** design class

For higher resolution the design class [here](#)

12. System Architecture

In this section, we will illustrate the architectural style chosen for the FAYDH application and define the structure, behavior, and views of the system. The architectural style is a critical factor in shaping the application's foundation, influencing how its components interact, communicate, and operate under varying conditions. Our primary objective is to establish a robust and organized structure that ensures seamless availability, reliable performance, and a fluid user experience. Given the multi-role nature of FAYDH—with donors, charities, drivers, and administrators interacting simultaneously—the architecture must support real-time updates, dynamic data processing, and efficient handling of concurrent operations. By clearly defining the system's structure, behavior, and views, we aim to create a framework that allows smooth coordination of all features, such as donation management, reservations, deliveries, and notifications, while maintaining flexibility for future expansion and scalability. This foundational approach will guide the development process and ensure that FAYDH remains responsive, maintainable, and capable of supporting its humanitarian objectives effectively.

12.1 MVVM Architecture style

The architecture chosen for the FAYDH application is the Model-View-ViewModel (MVVM) design pattern, which is a modern and efficient architecture commonly used in complex, data-driven, and real-time applications. MVVM organizes the system into three core components that separate responsibilities clearly and promote clean, maintainable programming practices.

The **Model** represents the core data and business logic of the application. It manages operations such as tracking donation posts, verifying reservation status, updating driver information, handling authentication, and maintaining user profiles. The Model is responsible for retrieving and storing data and responding to state changes triggered by system rules.

The **View** displays the user interface for donors, charities, drivers, and administrators. It presents information such as available donation posts, reservation status, delivery progress, notifications, and dashboard summaries. The View remains focused purely on UI layout and visual presentation without containing business logic.

The **ViewModel** acts as the intermediary between the Model and the View. It exposes data from the Model in a way that the View can easily display, and it handles user actions such as making a reservation, generating verification codes, or approving driver requests. In MVVM, the View observes changes in the ViewModel, enabling automatic UI updates without manual refreshes—an essential capability for FAYDH's real-time features.

MVVM is particularly beneficial for the FAYDH application because different team members can work on separate layers in parallel. Developers can improve backend logic in the Model without affecting UI layout, while designers and frontend developers can adjust the View independently. Meanwhile, the ViewModel ensures the two layers interact smoothly. MVVM architecture was selected because it can promote the non-functional requirements for the system as follows:

- **Maintainability:**

MVVM makes the application easier to maintain by separating logic into independent layers. Each ViewModel contains focused business logic, making it simple to update donation rules, modify reservation logic, or adjust driver-tracking behavior without impacting UI components. This maintainability is crucial for a multi-role humanitarian

system like FAYDH.

- **Scalability:**

MVVM supports modular expansion. As the number of users increases—or as new features such as additional donation types or new verification methods are introduced—the system can grow without restructuring the entire codebase. New ViewModels and Models can be added easily to support additional modules.

- **Extensibility:**

Clear separation of concerns allows the FAYDH platform to evolve over time. Future enhancements like AI-based matching, advanced analytics for charities, or expanded delivery options can be integrated by introducing new Models and ViewModels with minimal disruption to existing views.

- **Testability:**

MVVM greatly improves testability because ViewModels contain the core system logic without being tied to the UI. This allows developers to test rules such as validating donation expiry, calculating rewards, assigning drivers, or sending notifications in isolation. Reliable testability ensures that FAYDH remains consistent and trustworthy as a humanitarian platform.

While MVVM offers numerous benefits, it also has challenges when applied to applications like FAYDH. For example, implementing MVVM can introduce additional complexity because it requires careful planning to structure Models, Views, and ViewModels correctly. It may also lead to a larger number of files, which can feel overwhelming for small teams. Additionally, developers unfamiliar with reactive programming or data binding may initially find MVVM more challenging to implement than simpler architectures. However, the advantages of clarity, scalability, and responsiveness outweigh these limitations for a large, real-time system like **FAYDH**. [9]

12.2 MVVM Architecture for Admin Website

In addition to the mobile application used by donors, charities, and drivers, the administrator interacts with FAYDH through a web-based platform that also follows the MVVM architectural style. Although the admin interface is delivered through a browser, the underlying structure remains consistent with the mobile application. The web version separates the interface (View), business logic (Model), and state-handling layer (ViewModel) just like the mobile app. This allows administrators to manage users, monitor donations, review reports, verify charities, and oversee system activity efficiently using a clean and structured architecture. By applying MVVM to the admin's web portal, FAYDH ensures unified behavior across both mobile and web environments, enabling real-time updates, consistent data handling, and seamless maintainability. It also ensures that any backend changes—such as modifying approval workflows or adjusting reporting mechanisms—will automatically reflect in both the app and the admin dashboard without redundant development effort.

12.3 Architecture Constraints

Several constraints influence the overall architecture of the FAYDH system. First, the platform must support real-time updates for donations, reservations, and deliveries, which imposes a constraint on how data flows between components. Second, strict security and privacy constraints require controlled access to user roles, encrypted communication, and safe handling of sensitive information. Third, the system must operate efficiently across different devices and unstable network conditions, creating performance constraints that limit how much processing can be done on the client side. Fourth, FAYDH must remain scalable as the number of donations, users, and concurrent operations increases, placing constraints on how data is structured and how modules communicate. Finally, maintainability constraints arise from the multi-role environment (donor, charity, driver, admin), which requires the architecture to isolate logic and avoid tightly coupled components to prevent errors when features evolve.[10]

12.4 MVVM Architecture compared to other Architectures

Compared to other architectural styles, MVVM provides clearer separation of concerns and better state management for modern applications. Unlike MVC, where controllers are often tightly coupled with views, MVVM isolates UI logic within the ViewModel, improving maintainability and testability. Compared to MVP, MVVM avoids overly complex presenters by managing UI state more efficiently, which enhances scalability as the system grows. While VIPER offers strong separation, it introduces significant complexity and development overhead that is unnecessary for applications like FAYDH. Similarly, Clean Architecture provides high abstraction but can be excessive for client-side logic. Overall, MVVM achieves a balance between simplicity, modularity, and testability, making it a practical choice for real-time, multi-role systems.[12]

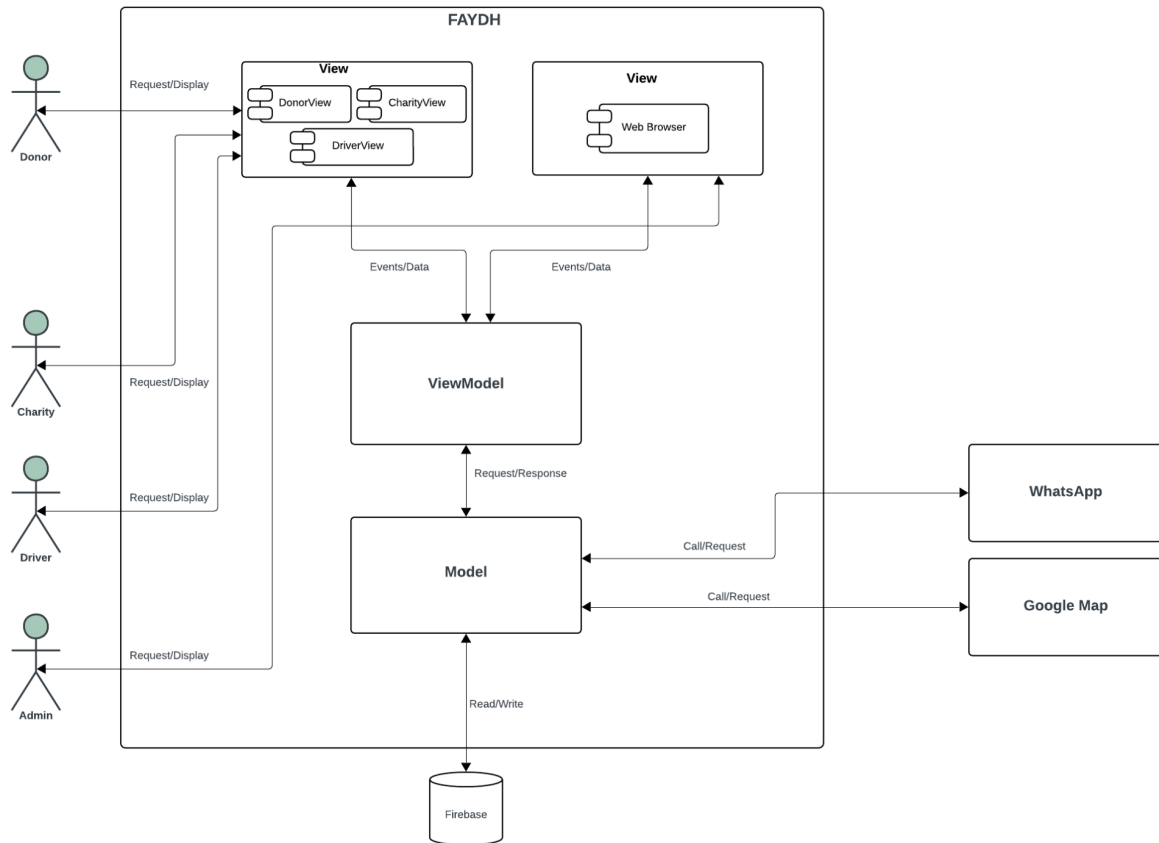


Figure 35. **FAYDH** Architecture Diagram

13. User Interface Mockup

Provide screenshots of the developed user interfaces, or mockups to illustrate the looks and feel of the system for critical scenarios.

13.1 Welcome Page & Sign In



Figure 36. Welcome and sign in page

13.2 Sign Up for charity

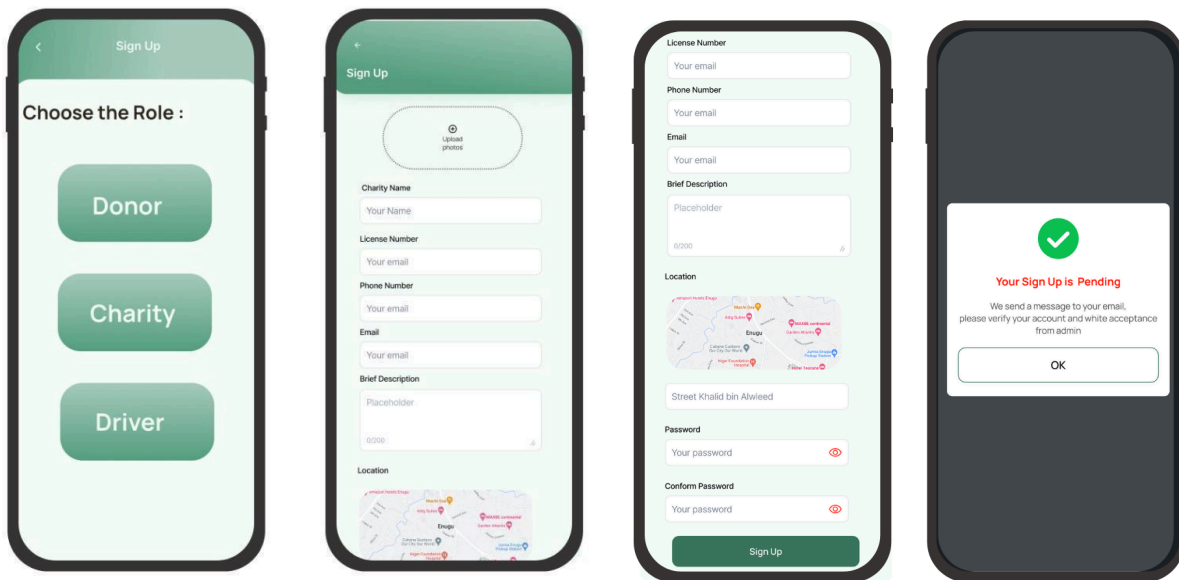


Figure 37. Sign up for charity

13.3 Create Donation Post

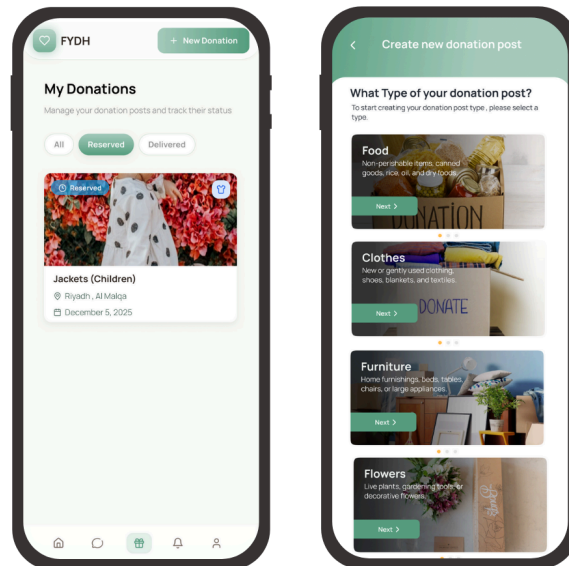


Figure 38. Donor Homepage, Donor clicks on New Donation to create a donation post.

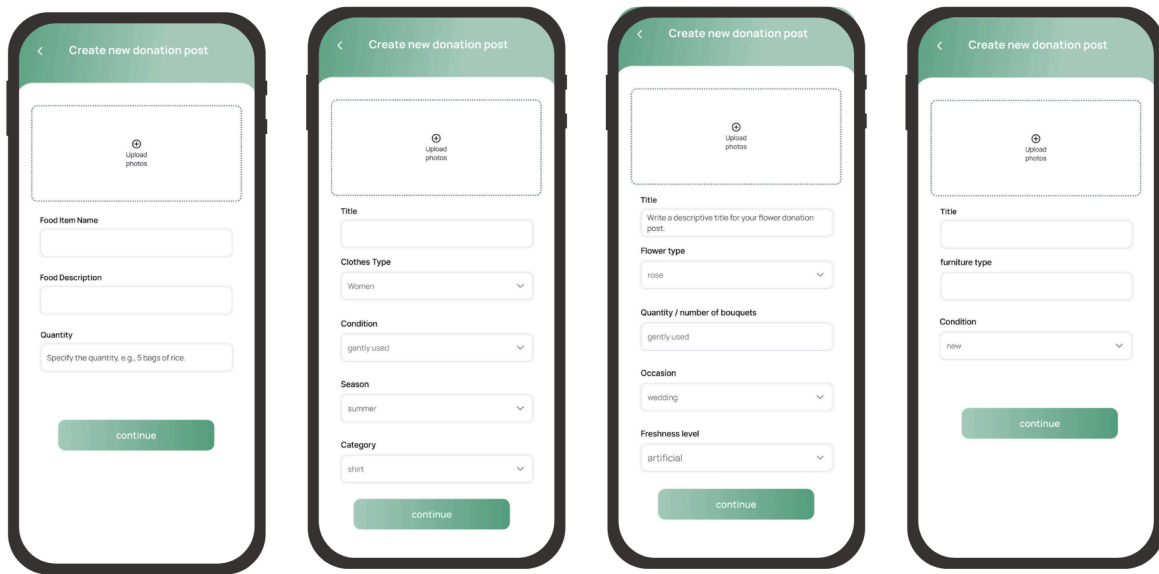


Figure 39. The donor fills in a form with the relevant type

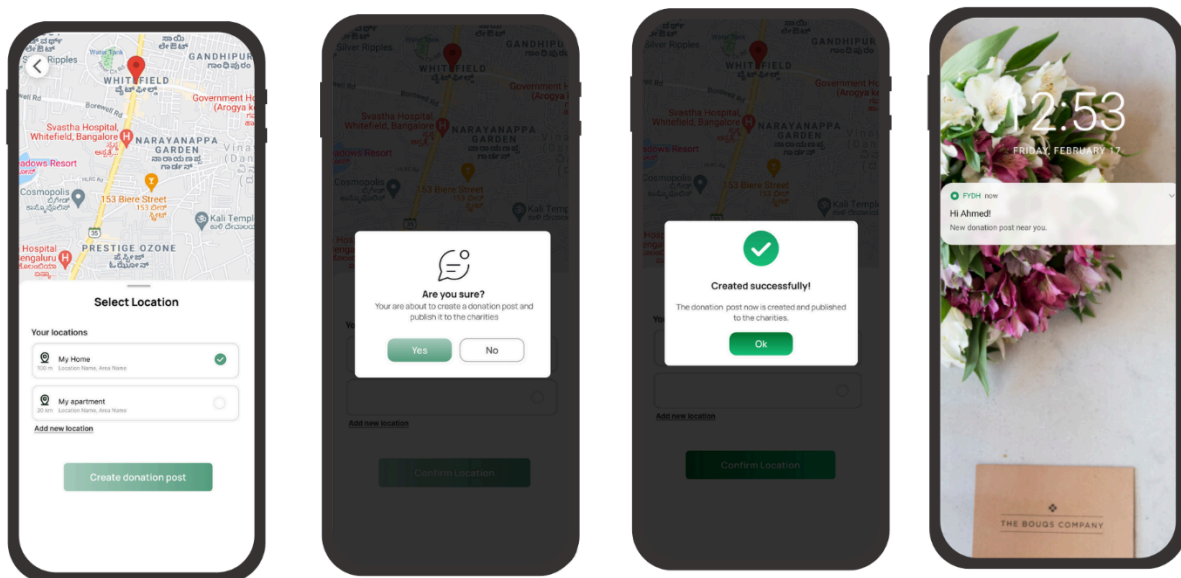


Figure 40. The donor can choose an existing location or add a new Location, after confirming the donation post the message will be received by who(Driver)

13.4 Join Chat

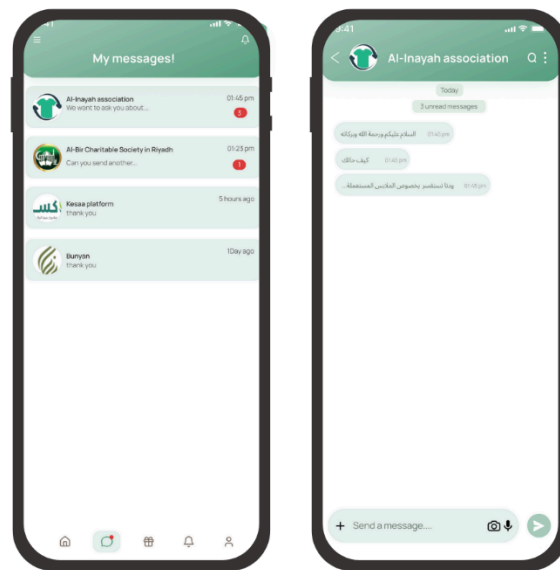


Figure 41. The donor can join a chat with the charity to give them more details about the donation.

13.5 Choose Available Pick-up time

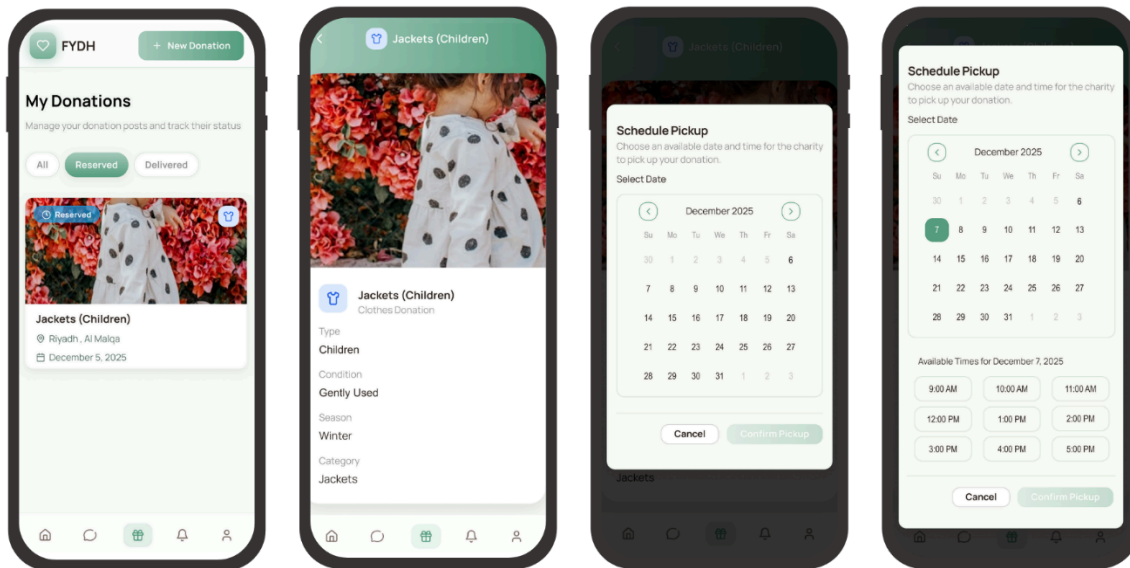


Figure 42. The donor can choose the available time for delivery pick up

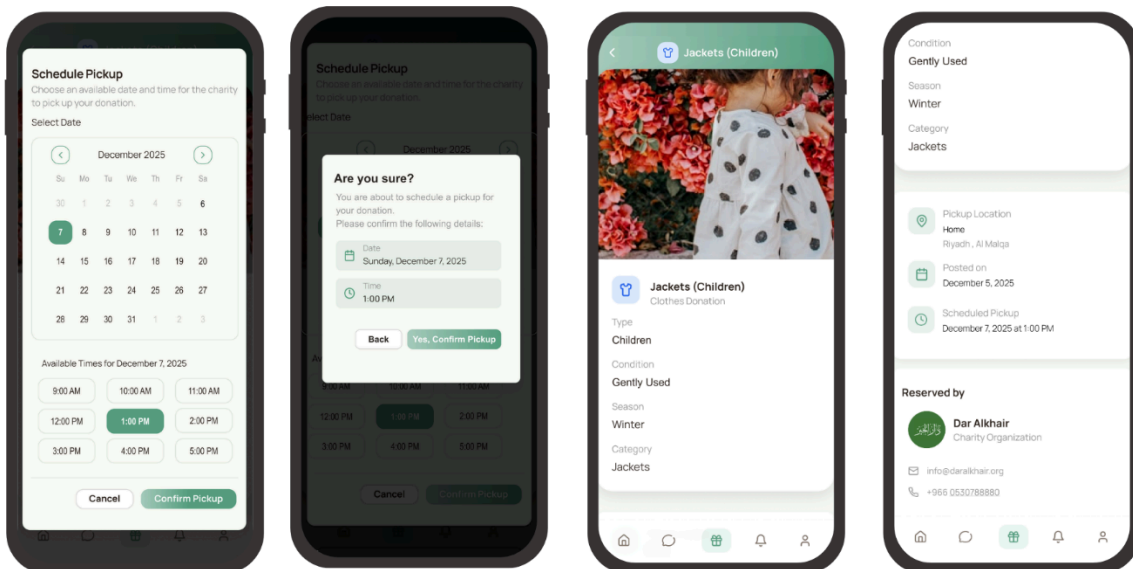


Figure 43. Confirmation steps and view the donation post details with scheduled pickup

13.6 Reserve donation post

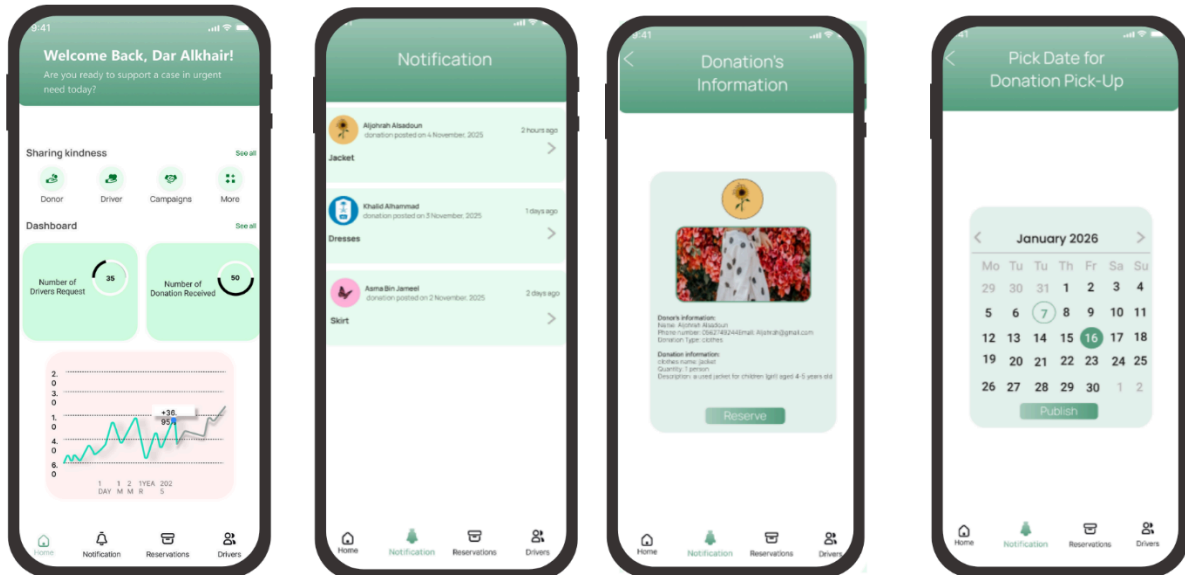


Figure 44. Charity Home page and receive notification for donation post, charity can view details and reserve a donation post and publish the time for donation pick- up

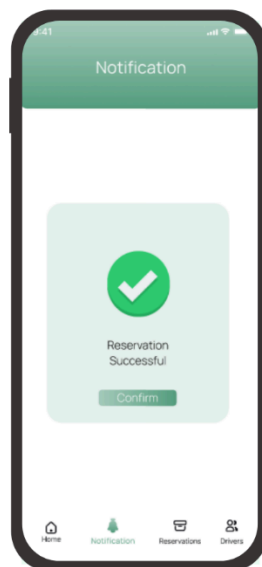


Figure 45. Confirmation message for reserved donation post

13.7 Cancel Donation Reservations

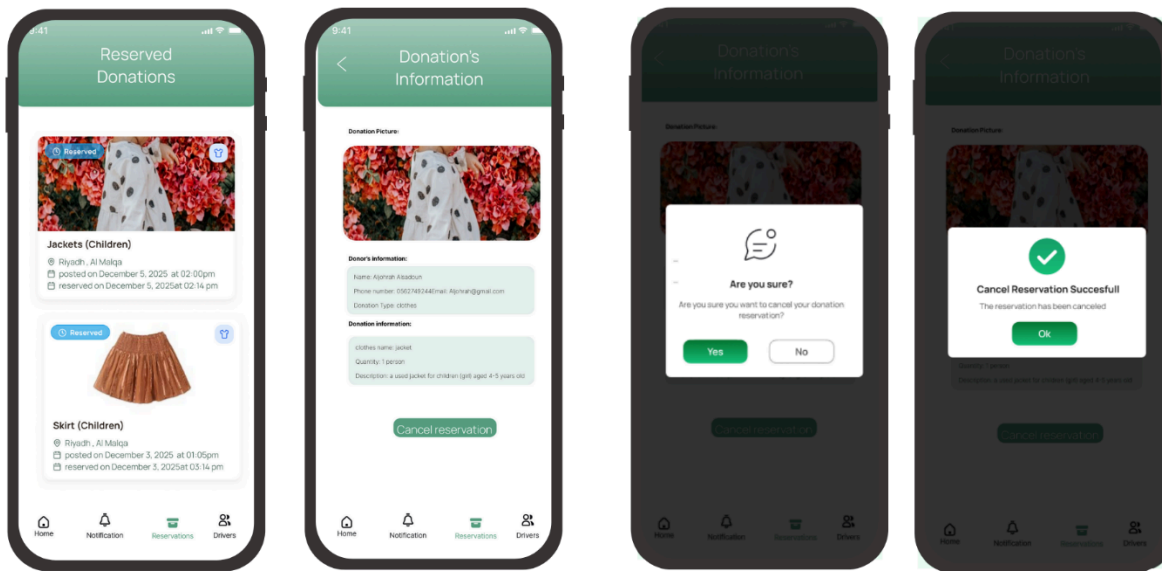


Figure 46. Charity can cancel reserved donation

13.8 Accept/Reject Drivers' Request

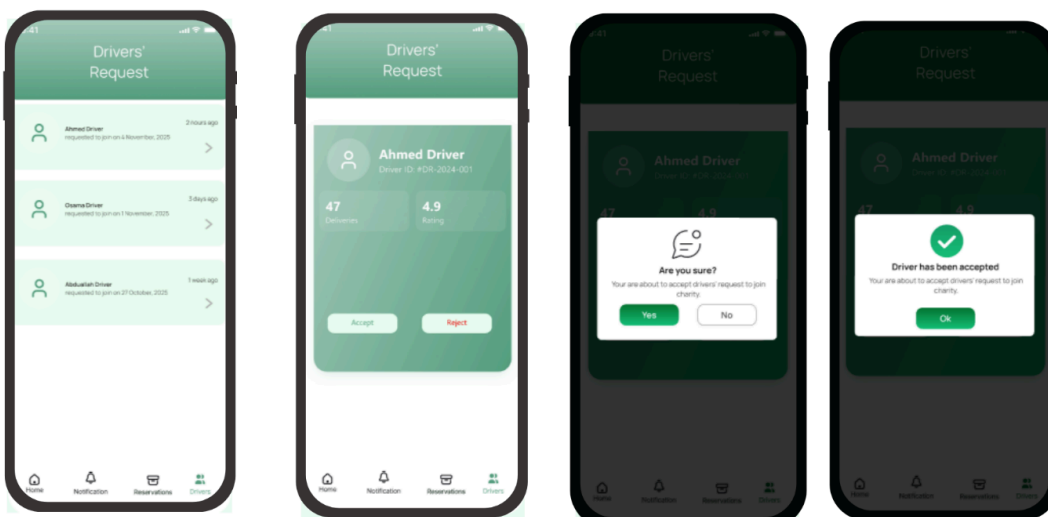


Figure 47. Charity view driver request and can accept or reject them

13.9 Update delivery status

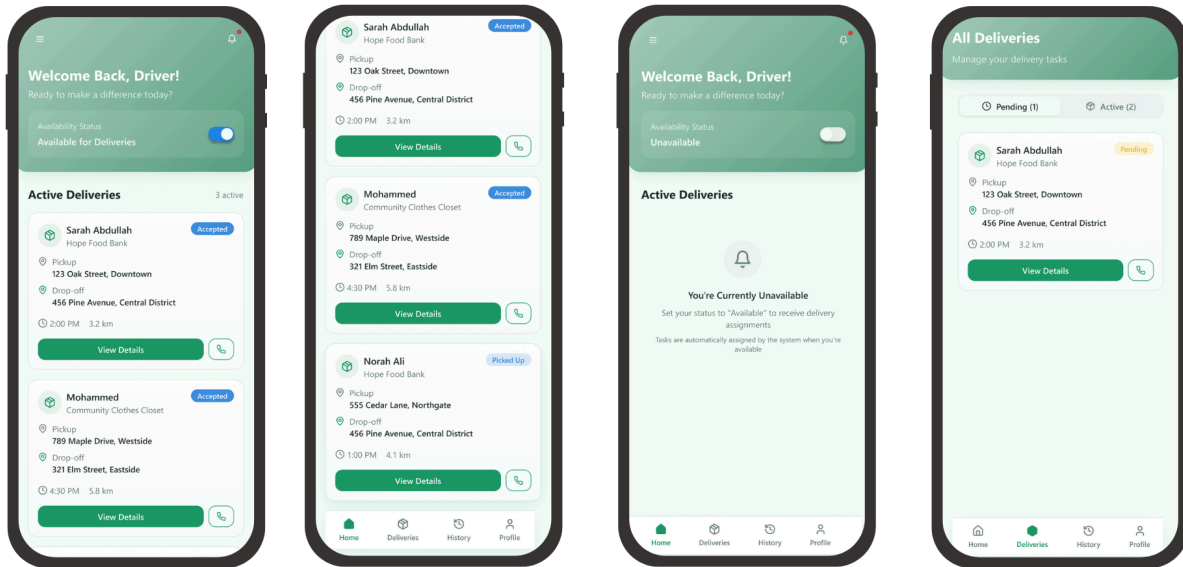


Figure 48. Driver mark himself to available to get assigned a delivery from the system

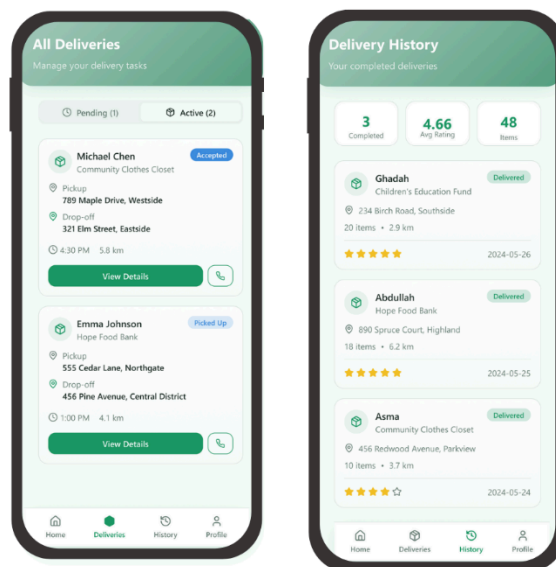


Figure 49. Driver can view active delivery and delivery history

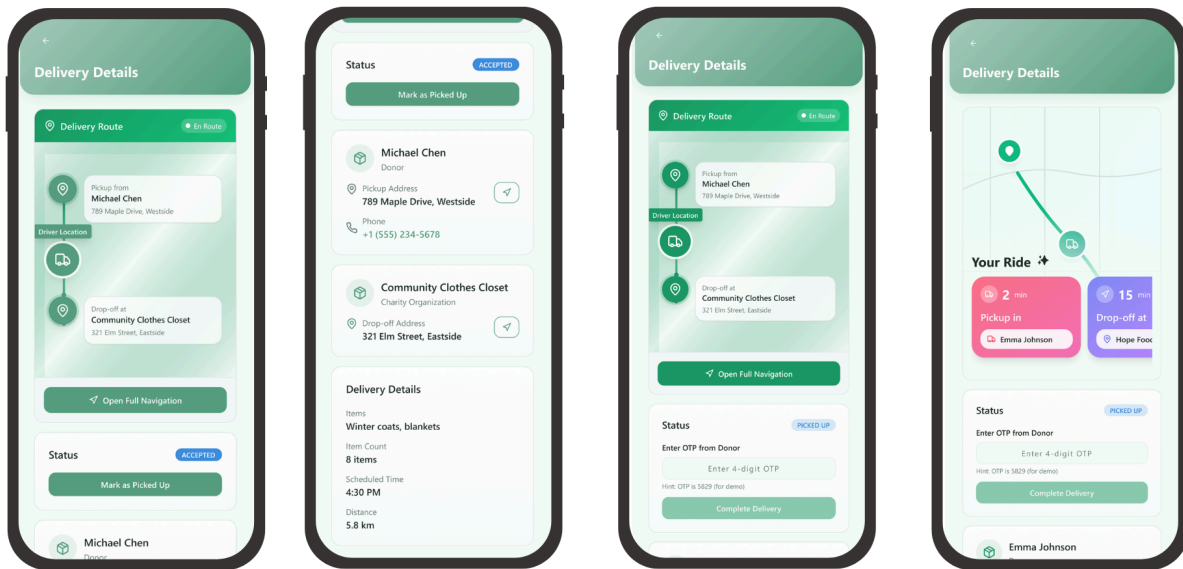


Figure 50. Driver view delivery details and enters the verification code

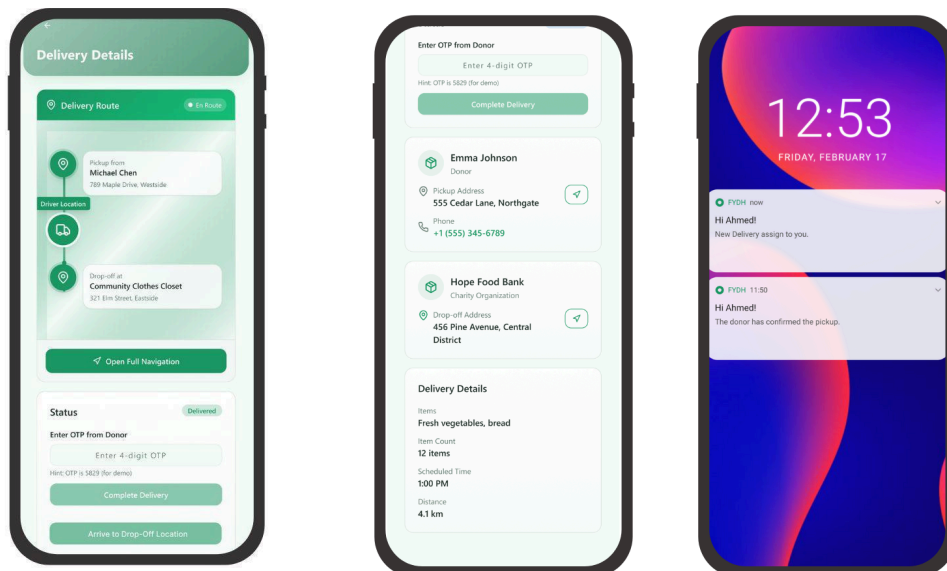


Figure 51. Driver view and navigate to Drop-off location, and receive notification.

13.10 Driver Request to join

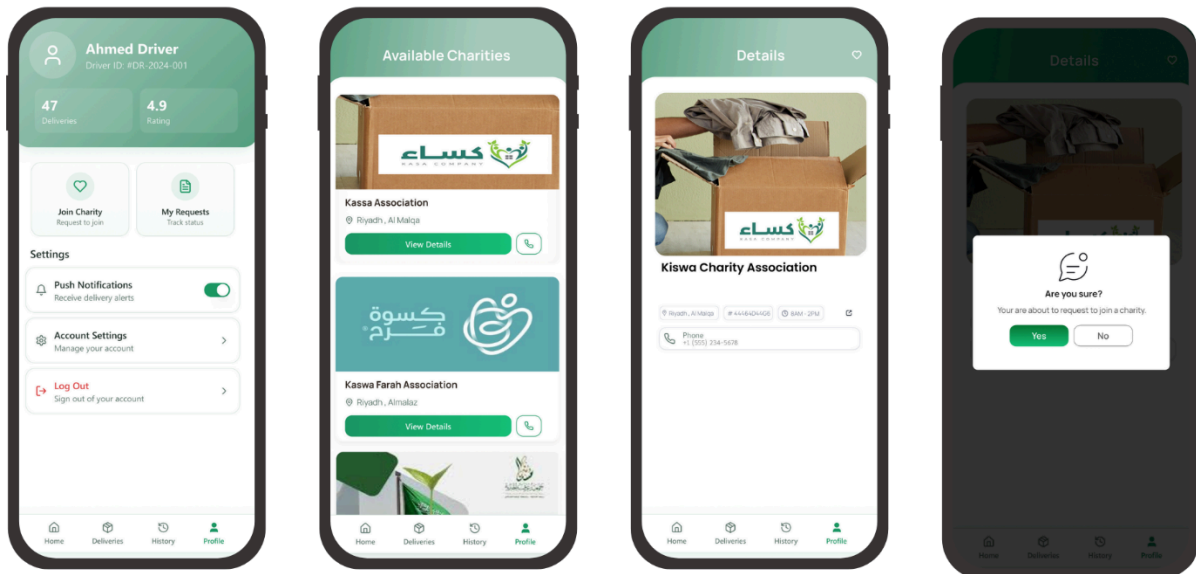


Figure 52. Driver request to join charity

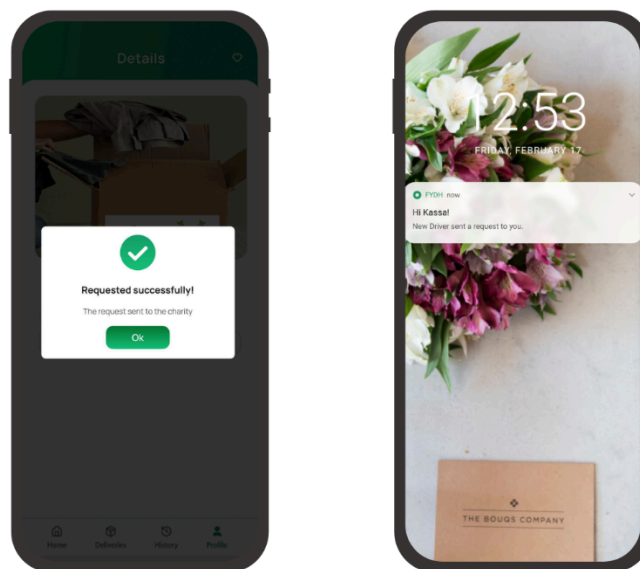


Figure 53. Conformation message for requested successfully

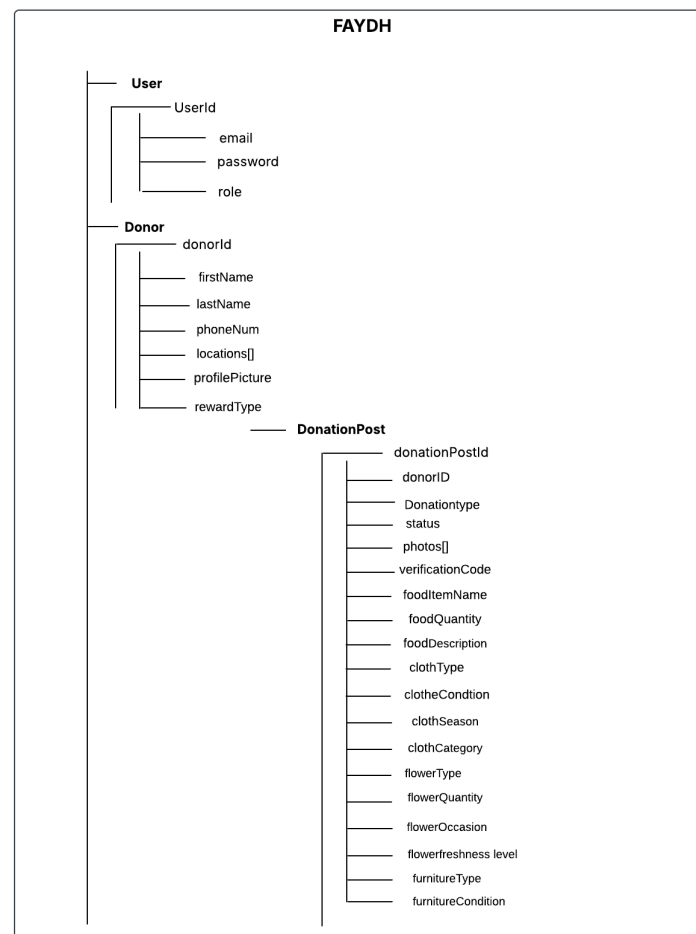
14. Database

Databases are structured collections of electronically stored information, managed by a Database Management System (DBMS), which facilitates data retrieval, updates, and efficient management. For **FAYDH**, a robust database is essential to track and manage system data seamlessly. To meet its dynamic and evolving requirements, Firebase Firestore, a NoSQL cloud-hosted database, has been chosen. Our prior experience with Firestore also influenced this choice, ensuring we can implement and optimize it effectively for **FAYDH**.

Firestore offers real-time synchronization, scalability, and flexibility, ensuring **FAYDH** can handle increasing data volumes and adapt to changing needs without structural complexities. Its document-oriented storage simplifies data organization, while its seamless integration with Firebase services like Authentication and Storage creates a unified backend ecosystem.

By leveraging Firebase Firestore, **FAYDH** gains a powerful and scalable backend solution that aligns with its goal of delivering a fast, responsive, and user-centric system. This decision ensures a streamlined development process while setting the foundation for future growth.[12]

14.1 Database Schema



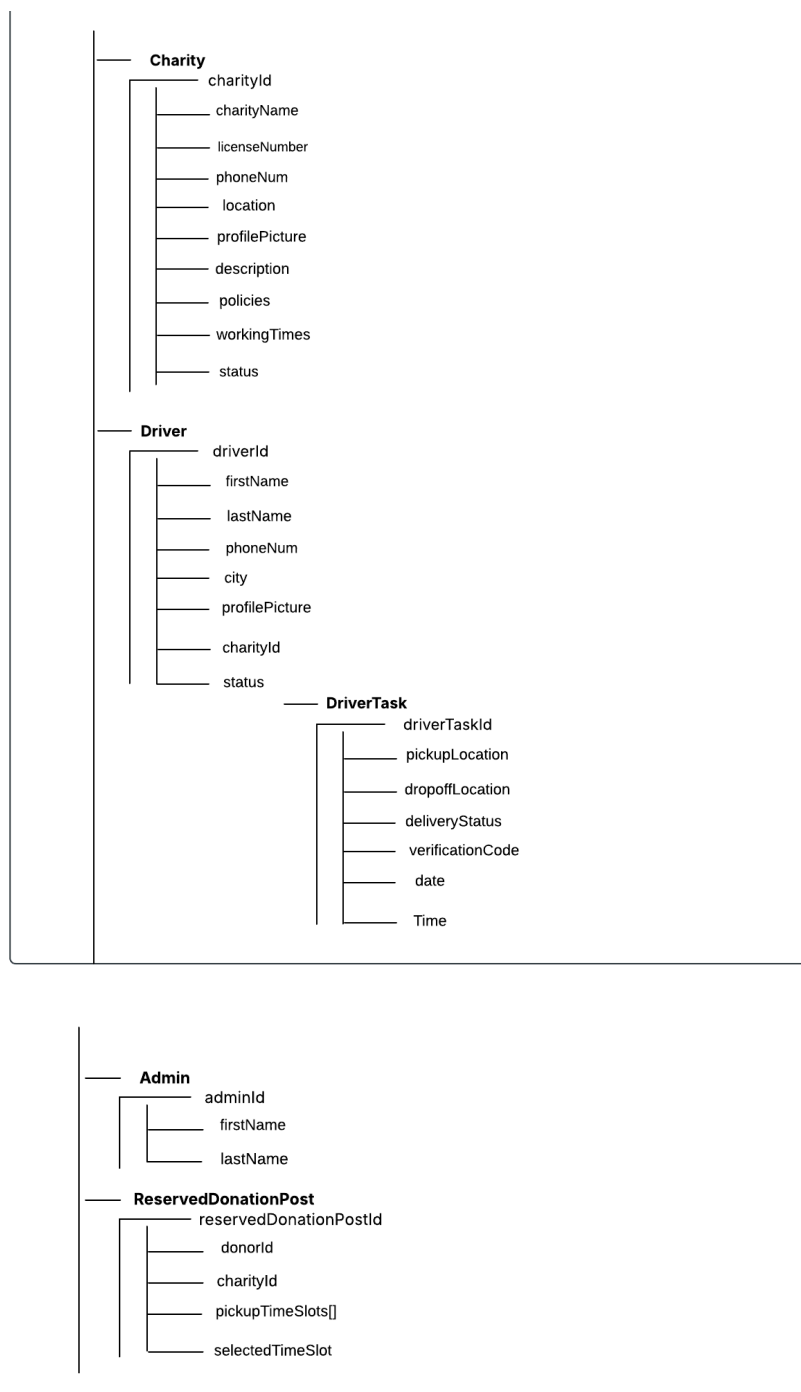


Figure 54. **FAYDH** database schema

15. Algorithms

15.1 delivery tasks Haversine

Delivery tasks are assigned to the nearest available driver relative to the donor's location. The system integrates **Google Maps services** to obtain real-time geographic coordinates for both donors and drivers. Distances are computed using the **Haversine formula**, which accounts for Earth's curvature for accurate GPS-based measurement.

Driver assignment uses a **weighted matching score** for each donor-driver pair:

Matching Score Formula:

$$\text{Score} = (\text{Distance Weight} \times \text{Distance}) + (\text{Urgency Weight} \times \text{Urgency} \times \text{TypeIndicator})$$

Where:

- **Distance** = distance between donor and driver (Haversine formula)
- **Urgency** = normalized urgency of the donation (0–1 scale)
- **Distance Weight (w_d)** = weight for distance (primary factor)
- **Urgency Weight (w_u)** = weight for urgency (applied only for food or flowers)
- **TypeIndicator** = 1 if the donation is food or flowers, 0 otherwise

The driver with the **lowest score** is assigned to the donation.

```

FUNCTION haversineDistance(lat1, lon1, lat2, lon2):
  R = 6371 # Earth's radius in km
  dLat = (lat2 - lat1) * (PI / 180)
  dLon = (lon2 - lon1) * (PI / 180)
  a = SIN(dLat/2)^2 + COS(lat1 * (PI / 180)) * COS(lat2 * (PI / 180)) * SIN(dLon/2)^2
  c = 2 * ATAN2(SQRT(a), SQRT(1 - a))
  distance = R * c
  RETURN distance

FUNCTION assignNearestDriver(taskLat, taskLon, taskType, taskUrgency, driversList, w_d, w_u):
  nearestDriver = NULL
  minScore = INFINITY
  FOR EACH driver IN driversList:
    IF driver.status == "AVAILABLE":
      distance = haversineDistance(taskLat, taskLon, driver.lat, driver.lon)
      # Conditional urgency only for food or flowers
      IF taskType IN ["food", "flowers"]:
        urgency = normalize(taskUrgency) # 0-1 scale
      ELSE:
        urgency = 0
      # Weighted matching score
      score = (w_d * distance) + (w_u * urgency)
      IF score < minScore:
        minScore = score
        nearestDriver = driver
  IF nearestDriver != NULL:
    nearestDriver.assignTask()
    RETURN nearestDriver
  ELSE:
    RETURN "No available drivers"

```

Figure 55. **FAYDH** Algorithm

16. Expected Deployment

The deployment diagram shows how the FAYDH system is distributed across three main environments: the mobile app for donors, charities, and drivers (Android), the admin web interface (Windows/Mac browsers), and the Firebase backend. All user devices communicate with Firebase through secure HTTPS to access authentication, Firestore database, storage for images, and real-time notifications. The system also integrates with external services such as Google Maps for navigation and WhatsApp through deep links, while the reset password will handle using email via SMTP. Each user role interacts with the same backend but through different interfaces designed for their specific tasks.

The application follows the MVVM architecture. The **View** displays the UI for each user role—for example, donors see screens for creating donation posts, charities see reservation and dashboard interfaces, drivers see delivery-task and map screens, and admins view approval and statistics pages. The **ViewModel** handles the logic behind these actions: donors submitting posts, charities reserving and scheduling pickups, drivers updating delivery status, and admins approving charity

accounts. It validates input, manages UI state, and communicates with Firebase. The **Model** stores the structured data for each role, such as donor profiles and posts, charity policies and reservations, driver tasks and locations, and system-wide metrics for admin. This separation ensures clean logic, real-time updates, and scalable performance for all users.

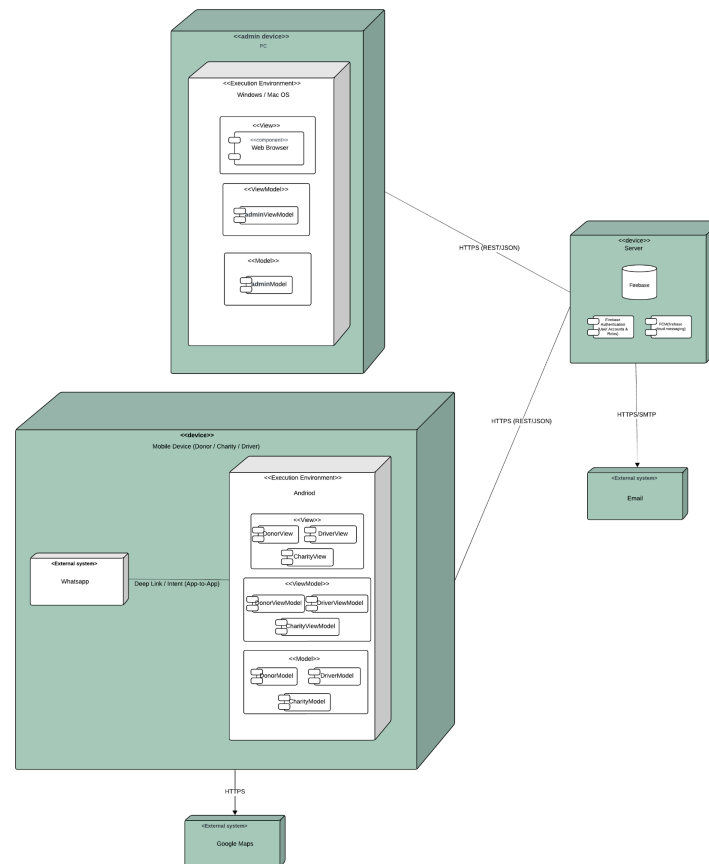


Figure 56. **FAYDH** deployment diagram

17. Test Scenario

17.1 Create donation post test cases

17.1.1 Invalid scenario

TC ID	TC1
Title	Create Donation Post - Missing Mandatory Photo Field
Description	To verify that the system prevents post submission when a mandatory field (Photos) is left empty, and displays the correct error message.
Precondition	1. The Donor is signed in to the.

Test Steps	1. Select "New Donation". 2. Select "Next" with donation type: Clothes. 3. Fill in all mandatory fields except the 'Photo(s)' field. 4. Click the "Continue" button.
Test Data	Clothes Type: "Women", Title: "Winter Jackets", Condition: "gently used", Season: "Winter", Category : "Jacket", Photo(s): Empty.
Expected Result	1. The system prevents the post from being submitted. 2. The missing 'Photo(s)' field is highlighted. 3. The system displays a clear error message ("Photo required"). 4. No notification is sent to charities.

Table 13. Create donation post test case

17.1.2 Valid scenario

TC ID	TC2
Title	Create Donation Post Successfully (Food Type)
Description	To verify that the Donor can successfully create and submit a new 'Food' donation post with valid data, and that the post is marked as "Available" and charities are notified.
Precondition	1. The Donor is signed in to the system.
Test Steps	1. Select "New Donation". 2. Select "Next" with donation type: Food. 3. Fill in all required fields in the Food form (Food Item Name, Quantity, Food Description, Photos). 4. Select "Continue". 5. Select an existing saved location. 6. Click the "Create donation post" button and confirm the submission message.
Test Data	Food Item Name: "Canned Goods", Quantity: 10, Food description: "I have a canned goods that is very good and it can be taken", Photos: 1 Valid JPEG file, Location: Saved Address A.
Expected Result	1. The system displays a success message ("created successfully"). 2. The new post appears in the Donor's "My Donations" list with the status "Available". 3. All charities receive an automatic notification about the new available post.

Table 14. Create donation post test case

17.2 Choose pickup available time test case

17.2.1 Valid scenario

TC ID	TC3
Title	User selects and schedules an available pickup date and time
Description	Verify that the donor can open a reserved donation post, select an available pickup date and time, confirm it, and successfully schedule the donation pickup.
Precondition	
Test Steps	1. Donor navigates to the Donations tab. 2. Donor selects the "Reserved" filter. 3. Donor opens a reserved donation post. 4. Donor taps "Schedule Pickup". 5. Donor selects a date from the displayed calendar. 6. Donor selects one available time slot. 7. Donor taps "Confirm Pickup". 8. Donor views the confirmation dialog with date & time details. 9. Donor taps "Yes, Confirm Pickup".
Test Data	Donation Post: "Jackets (Children)" Available Date: December 7, 2025 Available Time: 1:00 PM
Expected Result	1.The system displays a final success message confirming the scheduled pickup. 2.The donors are returned to the donation post details page with the updated scheduled pickup time.

Table 15.Choose pickup available time test case

17.3 Join chat test cases

17.3.1 Valid scenario

TC ID	TC4
Title	Join Chat Successfully

Description	This test case aims to test how the system will behave when the donor joins a valid chat room after the charity starts the chat.
Precondition	1-The charity is signed into the system. 2-The charity Start chat with donors for a reserved donation post. 3-The donor is signed into the system. 4-The donor receives a chat notification from the system. 5-A valid chat session exists and is active for this donation post.
Test Steps	1-Donor navigates to the “Chat” tab. 2-Donor selects the specific chat from list of chat sessions to join it
Test Data	None
Expected Result	The system opens the chat session and displays the chat screen in real time.

Table 16.Join chat test case

17.3.2 Invalid scenario

TC ID	TC5
Title	Join Chat When Donation Post Is Delivered
Description	This test case aims to test how the system will behave when the donor tries to join a chat after the donation post status has changed to “Delivered”.
Precondition	1.The charity is signed into the system. 2.The charity starts to chat with donors for a reserved donation post. 3.The donor is signed in to the system. 4.The donor receives a chat notification from the system. 5.A chat session exists for the donation post. 6.The related donation post status is turned to Delivered.
Test Steps	1.Donor navigates to the “Chat” tab. 2.The donor selects the chat that relates to a donation that has been delivered.
Test Data	None
Expected Result	1.The system closes the chat and displays a message:” the chat is no longer available.” 2.The donor cannot send new messages to the charity.

Table 17.Join chat test case

17.4 Reserve donation post test cases

17.4.1 Valid scenario

TC ID	TC6
Title	Reserve a donation post successfully.
Description	This test case verifies that a charity can successfully reserve an available donation post, select delivery time slots, and receive confirmation.
Precondition	1. Charity is signed in. 2. Charity has opened "View donation post details" for an available post.
Test Steps	1. Charity clicks "Reserve Post" 2. Charity selects available time slots using "Choose available time slots for delivery" 3. Charity confirms the reservation. 6. Charity views successful reservation message.
Test Data	Donation Post: "Jackets (Children)" Available Date: December 7, 2025 Available Time: 1:00 PM
Expected Result	The post status becomes "Reserved", it disappears from other charity accounts, a confirmation message appears saying "Reservation completed successfully.", and the donor receives a reservation notification.

Table 18. Reserve donation post test case

17.4.2 Invalid scenario

TC ID	TC8
Title	Attempt to reserve a donation post that is already reserved
Description	This test case verifies that the system prevents a charity from reserving a post that was already reserved by another charity.
Precondition	1. Charity is signed in. 2. Charity has opened "View donation post details".

	3. Donation post status = Reserved by another charity.
Test Steps	1. Charity clicks “Reserve Post”. 2. Charity sees an error message indicating the post is already reserved.
Test Data	Donation Post: “Jackets (Children)” Available Date: December 7, 2025 Available Time: 1:00 PM
Expected Result	The system displays “This donation post has already been reserved by another charity”, the reservation process stops, and the post remains unavailable.

Table 19.Reserve donation post test case

17.5 Cancel donation reservation post

17.5.1 Valid scenario

TC ID	TC7
Title	Cancel a reserved donation post successfully
Description	This test case verifies that a charity can successfully cancel a previously reserved donation post and that the system updates the post status accordingly
Precondition	1. The Charity is signed in. 2. The charity has opened the “View reservation donation post details” page. 3. Donation post is currently Reserved by this charity
Test Steps	1. Charity selects “Cancel Reservation”. 2. Charity confirms the cancellation. 3. Charity views the success message.
Test Data	Donation Post: “Jackets (Children)” Available Date: December 7, 2025 Available Time: 1:00 PM
Expected Result	The post status changes to “Available”, the post is removed from the charity’s My Reservations, the donor receives a cancellation notification, and the charity sees the message “Reservation canceled successfully.

Table 20.Cancel donation reservation post test case

17.5.2 Invalid scenario

TC ID	TC8
Title	Fail to cancel reservation due to inability to load donation post details
Description	This test case verifies that cancellation fails when the system cannot load the reserved donation post details.

Precondition	1. Charity is signed in. 2. Charity attempts to open “View reservation donation post details”. 3. System fails to retrieve donation post information
Test Steps	1. Charity selects “Cancel Reservation”. 2. Charity sees an error message indicating the post details cannot be loaded.
Test Data	Donation Post: “Jackets (Children)” Available Date: December 7, 2025 Available Time: 1:00 PM
Expected Result	The system displays the message “Unable to load donation post details. Please try again later.”, the post remains in Reserved status, no changes are saved, and the donor does not receive any notification.

Table 21. Cancel donation reservation post test case

17.6 view available donation post details

17.6.1 valid scenario

TC ID	TC9
Title	View Available Donation Post Details Successfully
Description	This test case aims to test how the system will behave when the charity selects an available .donation post to view its details

Precondition	1-The charity is signed into the system. 2-There are available donation posts.
Test Steps	1-Charity navigates to the notification tab 2-Select donation post from list of available donation posts.
Test Data	None
Expected Result	The system retrieves and displays all details of the selected donation post, including donor name, phone number, email, donation type, and donation form.

Table 22.View available donation post details test case

17.6.2 Invalid scenario

TC ID	TC10
Title	Donor deletes the post while the charity is viewing its details
Description	This test case aims to stop viewing the donation post details,when donor deletes a donation post while the Charity is viewing the donation post details.
Precondition	1-The charity is signed into the system. 2-There are available donation posts.
Test Steps	1-Charity navigates to the notification tab. 2-Select donation post from list of available donation posts. 3-The donor deletes the donation post currently being viewed by the charity.
Test Data	None
Expected Result	The system will display a message: "This donation post is no longer available." Then the system redirects the charity back to the donation posts list.

Table 23.View available donation post details test case

17.6.3 Invalid scenario

TC ID	TC11
Title	Another charity reserves the post while it's being viewed
Description	This test case aims to stop the charity from viewing the donation post details ,when another charity reserved the same donation post at the same time while viewing it .

Precondition	1-The charity is signed into the system. 2-There is an available donation posts.
Test Steps	1-Charity navigates to the notification tab. 2-Select donation post from list of available donation posts to view its details. 3-Another charity reserved the same donation post currently being viewed by the first charity.
Test Data	None
Expected Result	The system will displays a message: “This donation post is no longer available.” Then the system redirects the charity back to the donation posts list.

Table 24.View available donation post details test case

17.6.4 Invalid scenario

TC ID	TC12
Title	Donation Post will expires while charity viewing its details.
Description	This test case aims to stop the charity from viewing the donation post details , when a donation post expires while the charity is viewing the donation post details.
Precondition	1-The charity is signed into the system. 2-There is an available donation posts.
Test Steps	1-Charity navigates to the notification tab. 2-Select donation post from list of available donation posts to view its details. 3-The donation post will expiring while its being viewed by the charity.
Test Data	None.
Expected Result	The system will display a message: “This donation post is no longer available.” Then the system redirects the charity back to the donation posts list.

Table 25.View available donation post details test case

17.7 Choose available time slots for delivery

17.7.1 valid scenario

TC ID	TC13
Title	Charity selects an available delivery date and time slots for the donation
Description	Verify that the charity can select an available delivery date and time while completing the reservation process.
Precondition	1. The charity is signed in. 2. The charity is currently in the reservation process for a donation post.
Test Steps	1. Navigate to the Donations Posts List. 2. Select a donation post to view its details. 3. Click Reserve to start the reservation process. 4. Select one or more available delivery dates from the calendar. 5. For each selected date, choose one or more available time slots. 6. Confirm the reservation.
Test Data	Delivery Dates: e.g., 2025-12-8, 2025-12-9, 2025-12-10 Time Slots per Date: 09:00–11:00, 11:00–13:00, 14:00–16:00
Expected Result	Charity can complete the reservation process

Table 26.Choose available time slots for delivery test case

17.7.2 invalid scenario

TC ID	TC14
Title	System prevents charity from selecting overlapping delivery time slots

Description	Verify that the system detects and rejects overlapping time slots when the charity selects multiple dates and time slots during the reservation process.
Precondition	1.The charity is signed in. 2. The charity is currently in the reservation process for a donation post.
Test Steps	1.Navigate to the Donations Posts List. 2.Select a donation post to view its details. 3.Click Reserve to start the reservation process. 4.Select one or more available delivery dates from the calendar. 5.For each selected date, choose one or more available time slots. 6.Confirm the reservation.
Test Data	Delivery Dates: e.g., 2025-12-8, 2025-12-10, 2025-12-11 Time Slots per Date: 09:00–11:00, 11:00–13:00, 14:00–16:00
Expected Result	1.The system detects the overlapping time slots. 2.The system displays a clear error message specifying the exact conflict: "The selected time slot (11:00–13:00) on 2025-12-11 overlaps with an existing slot (10:00–12:00). Please choose a different time slot."

Table 27.Choose available time slots for delivery test case

17.8 Request to join charity

17.8.1 Valid scenario

TC ID	TC15
Title	Request to Join Charity Successfully
Description	To verify that a Driver can successfully send a join request to a Charity, and that the request is recorded in the system and the Charity is notified.
Precondition	1. The Driver is signed in. 2. The Driver has not previously sent a request to the target Charity,
Test Steps	1. The Driver views the list of available charities. 2. The Driver selects "View Details" for the target Charity. 3. The Driver clicks the "Request to join" button. 4. The Driver confirms the confirmation message.
Test Data	Driver ID: #DR-2024-001, Charity Name: Kiswa Charity Association.
Expected Result	1. The system displays a success message ("Requested successfully").

	2. The system saves the request in the database with the status "Pending". 3. The system sends an automatic notification to the Charity about the new driver rst .
--	---

Table 28.Request to join test case

17.8.1 Invalid scenario

TC ID	TC16
Title	Request to Join Charity - Already Pending
Description	To verify the system's behavior when a Driver attempts to send a second join request to a Charity while their first request is still awaiting approval (pending).
Precondition	1. The Driver is signed in. 2. The Driver has an existing, pending request to the target Charity.
Test Steps	1. The Driver views the list of available charities. 2. The Driver selects "View Details" with the target Charity again. 3. The Driver attempts to click the "Request to join" button.
Test Data	Driver ID: #DR-2024-001, Charity Name: Kiswa Charity Association.Current Status: Pending.
Expected Result	1. The system prevents the submission of the new request. 2. The system displays the specific error message: "Request already pending. Please wait for approval." 3. No new record or notification is generated.

Table 29.Request to join test case

17.9 Update delivery status

17.9.1 Valid scenario

TC ID	TC17
Title	Update Delivery Status
Description	This test case aims to verify how the system behaves when a driver updates the delivery status through all required stages (Picked Up → Delivered).
Precondition	1- The driver is signed into the system. 2- The driver has an assigned active delivery.
Test Steps	1- The driver signs in and opens the assigned delivery. 2- The driver views the delivery details.

	3- The driver arrives at the donor location and enters the verification code. 4- The driver submits the correct code. 5- The driver arrives at the charity and taps "Delivered."
Test Data	Verification Code: "5482"
Expected Result	1- The system displays delivery details correctly. 2- Verification code is accepted. 3- Status changes from Active → Picked Up . 4- When "Delivered" is tapped, status changes to Delivered . 5- Confirmation message appears: "Delivery completed successfully." 6- Delivery is removed from the driver's active list.

Table 30.Update delivery status test case

17.9.2 Invalid scenario

TC ID	TC18
Title	Invalid Attempt to Mark Delivery Without Pick-Up
Description	Verify that the system prevents the driver from marking a delivery as "Delivered" before completing the pick-up verification.
Precondition	1- The driver is logged into the application. 2- Delivery is still in Active status (pick-up not verified).
Test Steps	1- The driver signs in and opens the delivery. 2- The driver views the delivery details. 3- The driver skips the donor verification step. 4- The driver taps "Delivered" at the charity location.
Test Data	Verification Code: None / Not entered
Expected Result	1- system displays a warning message: "You must pick up the delivery before marking it as Delivered." 2- The System redirects the driver back to the pick-up verification step. 3- Delivery status remains Active (no changes applied).

Table 31.Update delivery status test case

TC ID	TC21
Title	Invalid password format
Description	This test ensures that the system validates the password field and prevents sign-up processes with an incorrect password format.
Precondition	-
Test Steps	1-The user enters the application. 2- the user clicks on the “New her? Join Us!” 3- The user choose charity role 4- The user enters the charity name, license number, Phone number, Email, password, profile picture (optional), Brief Description, and Location. 5- The user clicks on “Sign Up”.
Test Data	charity name:”AL-bar” license number: “39864750152” Phone number: “0538746513” Email: “ <u>Albar@gmail.com</u> ” Brief Description: “jjjjjjjjjjjjjjjjjjjjjjjjjjjjjjjjj” Location:” Riyadh, Almulab street “ Password: Al1234@ Confirm password :Al1234@
Expected Result	The system will prevent the user from completing the sign up process and display an error message indicating that the password is not valid.

86

Test Data	charity name:"AL-bar" license number: "39864750152" Phone number: "0538746513" Email: " <u>Albar@gmail.com</u> " Brief Description: "jjjjjjjjjjjjjjjjjjjjjjjjjjjjjjjjj" Location:" Riyadh, Almulab street " Password: Al1234@ Confirm password :Al1234@
Expected Result	The system will prevent the user from completing the sign up process and display an error message indicating that the Licence number already exists .

Table 35.Sign up test case

18. Project Status

This section presents the project status, providing insights into the project's current state and the various stages it navigated through. It represents the current state or condition of a project at a specific point in time. It encompasses an assessment of the progress and overall health of the project. Understanding the project status is crucial for various stakeholders, including team members, project managers, sponsors, and other relevant parties. It provides a snapshot of where the project stands in relation to its goals and objectives.

Phase	Status	Start Date	End Date	Description
GP1				
Phase 1	Completed	14/9/2025	24/9/2025	The team conducted a thorough

Project Proposal				domain analysis of similar charitable platforms to identify key risks and constraints. Based on these findings, a structured project plan with clear milestones was developed, along with a basic quality assurance framework. Although predicting risks was challenging due to system complexity and multiple user roles, collaboration helped reduce uncertainties. This phase provided a solid foundation and a clear roadmap for the next stages of development.
Phase 2 Requirement analysis	Completed	26/9/2025	15/10/2025	In this phase, the team focused on eliciting and defining the functional and non-functional requirements of the FAYDH application. Use case diagrams, analysis classes, and interaction diagrams were developed to clarify system behavior and form the foundation for later design work. The team faced challenges due to differing viewpoints, the complexity of coordinating requirements for four user roles, and the effort involved in creating accurate analysis diagrams. Gathering requirements through interviews and observations also required significant time. Through collaboration and refinement, the team produced a clear and accurate analysis that effectively guided the next project stages.
Phase 3 System Design and Architecture	Completed	16/10/2025	10/12/2025	In this phase, the team progressed to the design stage, creating detailed design classes and defining the system's overall architectural structure. User interface mock-ups were developed to visualize the layout and interactions for all user roles on the FAYDH platform. Algorithms were designed to support core functions such as donation processing,

				reservation management, and driver assignment. The team also built the database schema to organize and manage the system's diverse data efficiently. Key challenges included choosing an appropriate architecture for multiple user roles with differing workflows and ensuring mock-ups balanced functionality with ease of use. The team also worked to keep the design aligned with project goals and avoid unnecessary complexity. Overall, this phase produced a complete blueprint to guide the implementation of the FAYDH application.
GP 2				
Phase 4 Implementation	Not started	TBD	TBD	In this phase, the team will implement Faydh application.
Phase 5 Testing	Not started	TBD	TBD	In this phase, the team will design the test cases to test the system to ensure we cover functional and nonfunctional requirements as specified previously. by using different testing techniques

Table 36. Project status

19. Future Work

While the **FAYDH** system provides a strong foundation for unified and transparent donation management, several opportunities exist for further enhancement and expansion:

Integration of Advanced Analytics:

Future iterations may incorporate predictive analytics to forecast donation patterns, identify urgent cases, and optimize distribution strategies for beneficiaries.

AI-Driven Matching Algorithms:

Intelligent algorithms could be implemented to automatically match donor preferences with beneficiary needs, improving efficiency and personalization.

Multilingual Features:

Expanding language support to include Arabic would broaden the platform's reach and improve inclusivity.

By advancing these areas, the **FAYDH** system can evolve into a more comprehensive, intelligent, and socially impactful platform capable of supporting the long-term digital transformation of charitable ecosystems.

20. Conclusion

The **FAYDH** system was conceived to address persistent challenges within the charitable sector by providing a unified digital platform that connects donors, beneficiaries, and charitable organizations with transparency and efficiency. Guided by a humanitarian vision rooted in solidarity and social responsibility, the system demonstrates how modern technology can enhance trust, reduce operational delays, and ensure that resources reach those most in need.

Through its streamlined donation workflows, real-time reporting mechanisms, and user-centric interface, the platform contributes to the digital transformation of charitable services while aligning with broader sustainable development goals. By consolidating multiple charitable entities within a single ecosystem, **FAYDH** strengthens collaboration among organizations and promotes wider community engagement. The system's architecture, analysis and design components, and database structure collectively support an integrated environment capable of improving decision-making, optimizing resource allocation, and maximizing social impact.

Ultimately, the **FAYDH** application represents a scalable model for technology-driven charitable work, highlighting the potential of digital platforms to address logistical and administrative challenges and to enhance transparency across all stages of the donation cycle.

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22. Appendix

Appendix 1

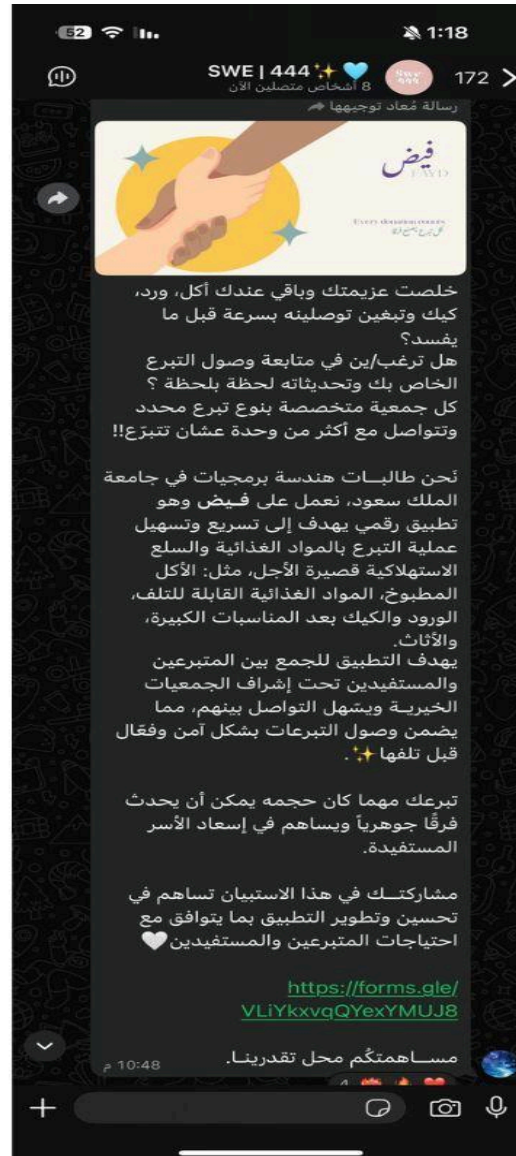
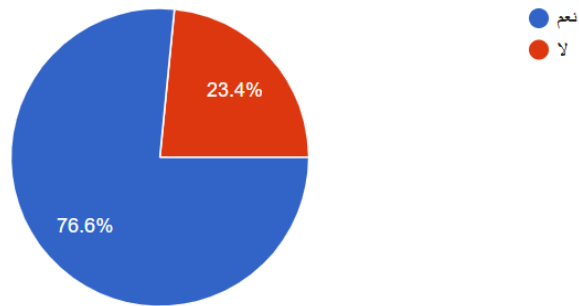


Figure 57. FAYDH Questionnaire

نسخ الرسم البياني

4. هل يهملك أن تعرف كيف تم استخدام تبرعك بالضبط؟

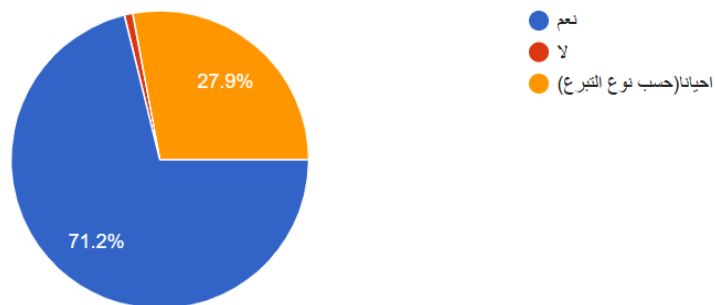
222 ردًا



نسخ الرسم البياني

5. هل تفضل وجود خدمة استلام التبرعات من المنزل عبر التوصيل؟

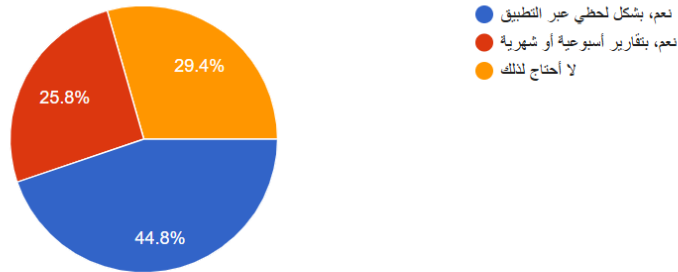
222 ردًا



نسخ الرسم البياني

6. هل ترغب باستلام إشعارات أو تقارير توضح أثر تبرعاتك؟

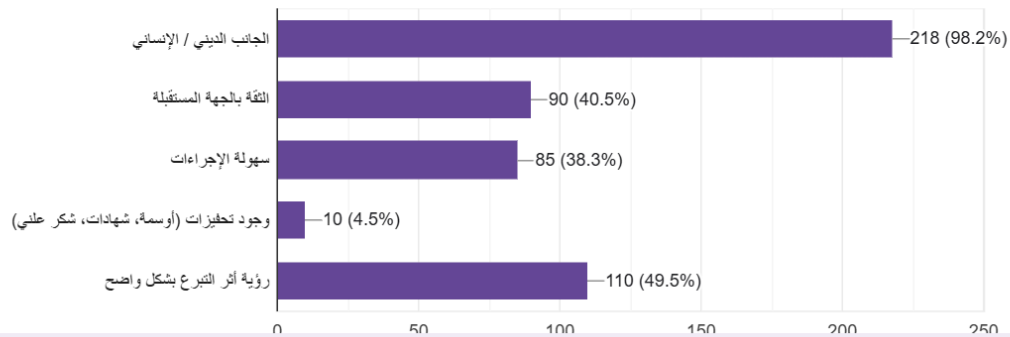
221 ردًا



نسخ الرسم البياني

7. ما الذي يحفزك أكثر للتبرع بانتظام؟

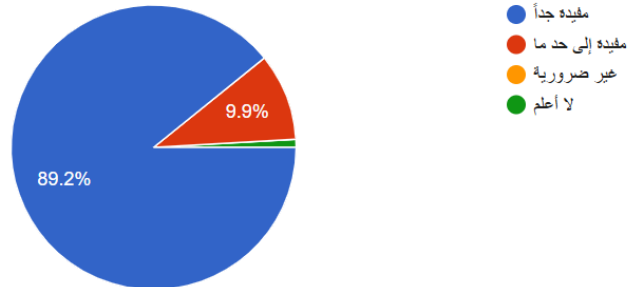
222 ردًا



نسخ الرسم البياني

8. ما رأيك في فكرة منصة ذكية لإدارة التبرعات تشمل (التبرع، التتبع، إدارة الحملات، التحفيز)؟

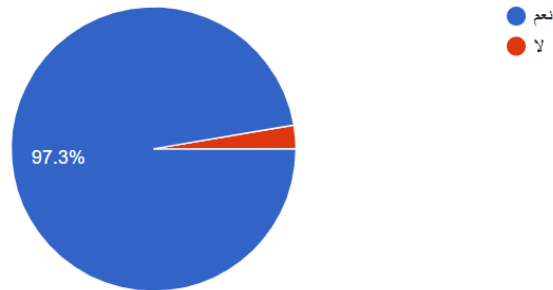
222 ردًا



نسخ الرسم البياني

9. هل تفضل أن يكون التطبيق ثنائي اللغة (عربي / إنجليزي)؟

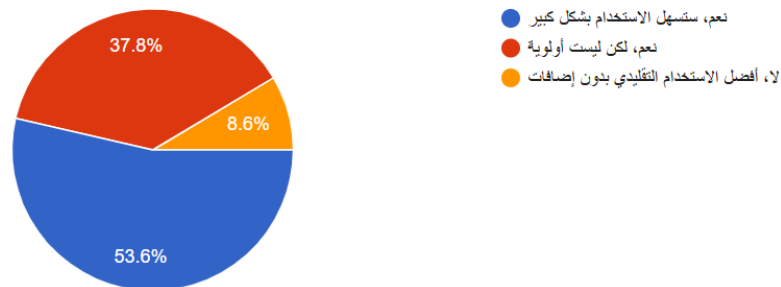
222 ردًا



نسخ الرسم البياني

10. هل تجد إضافة تقنيات مثل المساعد الذكي (Chatbot) أو التوصيات حسب اهتمامات تبرعاتك ميزة مفيدة؟

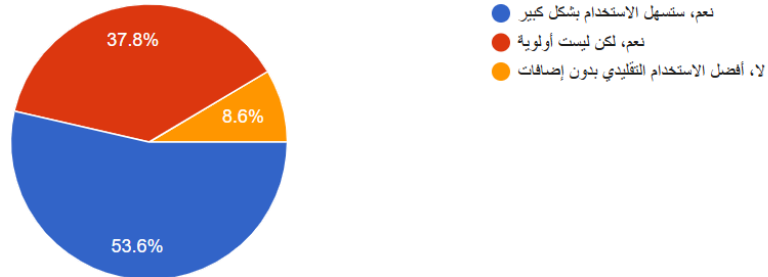
222 ردًا



نسخ الرسم البياني

10. هل تجد إضافة تقنيات مثل المساعد الذكي (Chatbot) أو التوصيات حسب اهتمامات تبرعاتك ميزة مفيدة؟

222 ردًا



11. ما اقتراحاتك لتحسين طرق التبرع، وهل تستخدم تطبيق أو منصة للتبرع، اذكرها؟

42 ردًا

سهولة شروط التبرع وسرعتها

الراجحي

ان يكون متوفر 24 ساعة لسرعه وسهولة نقل التبرعات للمحتاجين مع مراعات جودة المواد المتبرع بها ونتمنى لكم التوفيق

ا

احسان

السرعة في استلام التبرعات خصوصا الاكل(عشان ما يخرب)

نشر الوعي اكثر بين الجيل الجديد / اتيبرع لمنصة احسان

الأمانة في توزيع التبرعات للمستحقين وان تكون في مكانها الصحيح وان لا تستخدم الا لمن هم أولى بها ويكون هناك متابعة حقيقية وفعلية للأشخاص الذي يقومون بتوزيع التبرعات لضمان وصولها لمن يستحقونها

استخدام تطبيق، دعم تحديد الموقع، استخدام طريقة التبرع

Figure 58. FAYDH Questionnaire Result

Appendix 3

أسئلة المقابلة

التعريف بنفسى

الأسئلة العامة

- ١- ممكن تعطوني لمحة عن الجمعية وأهدافها الرئيسية؟
 - ٢- كيف يتم حالياً استقبال التبرعات والتواصل مع المستفيدين؟
 - ٣- مين هم الأطراف التي غالباً تتعاملون معهم (مكبرعين، مستفيدين، مندوبين... الخ)؟
2. العمليات الحالية والمشاكل
- وش أبرز التحديات التي تواجهكم في إدارة التبرعات حالياً؟
 - هل تواجهون صعوبات في توثيق ومتابعة وصول التبرعات للمستفيدين؟
 - كيف يتم حالياً اختيار أو تعيين المندوبين وتوزيع المهام عليهم؟
 - هل في مشاكل متكررة مع المكبرعين أو المستفيدين (مثل التواصل، المصادقية، تأخير التوصيل)؟
- كيف يتم فرز التبرعات عندكم داخل الجمعية هل تفصلون كل تبرع لحال ولا تصون لهم فرز بناء على وش
3. الجانب التقني والرقمي
- هل تستخدمون حالياً أي نظام أو تطبيق لإدارة التبرعات؟ ولو نعم، وش عيوبه؟
 - وش المميزات التي تكتنون وجودها في منصة أو تطبيق سهل صمكم؟
 - هل تحتاجون خاصية لتتبع الطلبات (من التبرع إلى وصولها للمستفيد)؟
4. الأمان والمصادقية
- كيف تضمنون حالياً وصول التبرعات لمستحقيها؟
 - هل عندكم حاجة لآلية تحقق من المكبرعين أو المستفيدين أو المندوبين؟
 - هل في تقارير أو إشادات معينة تحبون تظهر للتبرع بعد تنفيذ التبرع؟
5. المتطلبات الإضافية
- هل تحتاجون إلى تقارير دورية (مالية أو إحصائية) من المنصة؟
 - هل ترون أن وجود نظام تقييم (للمندوب أو للمستفيد أو للتبرع) مفيد؟
 - ما القنوات التي تفضلون أن يكون فيها إشعارات (داخل التطبيق، SMS، إيميل)؟
 - هل تحبون أن التطبيق يدعم أكثر من لغة (مثلاً: عربي/إنجليزي)؟
6. التطلعات المستقبلية
- لو وفرنا لكم منصة مثالية، وش أول 3 مشاكل تتوقعون تحظى من صمكم؟
 - هل في خطط مستقبلية للجمعية تحبون أن التطبيق يدعمها (مثلاً: التوسع الجغرافي، أنواع تبرعات جديدة)؟

Figure 59. FAYDH Interview Question

